Static Inequality Indices and Baseline-Independent Dynamic Extensions

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Abstract

We characterize which static inequality rankings admit baseline-independent dynamic extensions and what form those extensions can take. Such an extension preserves the maintained static judgment: a transition is weakly improving exactly when its final distribution is ranked weakly above its initial distribution. We prove that a continuous ordering on a convex domain has an affine representation when each feasible displacement is evaluated independently of the baseline from which it occurs. For a fixed population and fixed arithmetic mean, a continuous, nonconstant, and Schur-concave ranking admits such an extension if and only if it is a strictly increasing transformation of a generalized-Gini functional. The corresponding relative result yields generalized log-Gini functionals on fixed-geometric-mean domains of strictly positive incomes. For each admissible ranking, we characterize all baseline-independent dynamic extensions and the subclass that preserves Dynamic Lorenz comparisons. The generalized-Gini difference is a linear member of this subclass; it is Dynamic Lorenz consistent when the rank weights are strictly decreasing. The results extend to variable-transformed aggregates through centered effects. With variable populations, independent endpoint replication links the finite-population representations, while centered-quantile continuity yields a common continuous concave weighting function on population shares and extends the dynamic characterization to endpoints of different sizes.

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1 Introduction

A static inequality index ranks distributions, whereas a dynamic index evaluates the distributive content of transitions between distributions. This paper asks which ordinal static rankings can be extended dynamically so that the evaluation of a transition depends on its rank-based effects, not on the initial distribution from which those effects arise. This baseline-independent benchmark isolates

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the distributive content of the transition itself: two feasible transitions with the same rank-based effect profile receive the same dynamic value, even when they begin from different distributions. The paper characterizes both the static rankings compatible with this benchmark and the dynamic indices that extend them.

Static-Change Consistency ensures that the dynamic and static evaluations answer the same basic question. A transition from $\mathbf{x}$ to $\mathbf{y}$ is weakly improving according to the dynamic index exactly when the static index ranks $\mathbf{y}$ weakly above $\mathbf{x}$. Thus the dynamic index extends, rather than replaces, the maintained static ranking.

Baseline independence adds that this judgment must depend only on the rank-based effect of the transition. The same feasible effect must be classified in the same way wherever it occurs. Together, these requirements make the evaluation of every feasible displacement independent of its location in the domain. Under the maintained regularity conditions, this yields the affine generalized-Gini structure.

Baseline independence is intentionally demanding. It is appropriate when the aim is to evaluate the distributive content of a rank-based effect profile separately from the initial inequality of the distribution on which that effect is implemented. The condition is not maintained as a universal requirement: if initial inequality is itself relevant to the evaluation of a reform, the same effect may reasonably receive different evaluations at different baselines.

Relation to Dynamic Lorenz dominance. Carbonell-Nicolau (2026) develops absolute and relative Dynamic Lorenz dominance as ordinal, index-free criteria for comparing the inequality-reducing content of transitions. That paper takes a dynamic preorder as the primitive object and recovers a static comparison of $\mathbf{x}$ and $\mathbf{y}$ by comparing the transition $(\mathbf{x}, \mathbf{y})$ with the null transition $(\mathbf{x}, \mathbf{x})$.

The present paper studies the analogous problem for indices, but in the opposite direction. It begins with a static index and asks when a baseline-independent dynamic index can reproduce its ranking of the endpoints. Dynamic Lorenz dominance is not the object to be represented. Rather, it provides an additional criterion for comparing dynamic evaluations: among the admissible extensions characterized here, some preserve Dynamic Lorenz comparisons and others do not.

The two papers use the same absolute and relative effect geometries and related population-replication principles but characterize different objects. The companion paper characterizes dynamic dominance relations, whereas the present paper characterizes the static rankings that admit baseline-independent dynamic extensions and the indices that provide those extensions.

Main results. The first result is an affine representation theorem. It shows that, when the evaluation of each feasible displacement is independent of the baseline from which it is implemented, a continuous and nonconstant ordering on a convex domain can be represented by the nonnegative values of a nonzero linear functional.

Applying this result to inequality rankings yields the main fixed-population characterizations. In identity coordinates, on a domain with a fixed arithmetic mean, a continuous, nonconstant, and Schur-concave static ranking admits a normalized, baseline-independent dynamic index satisfying Static-Change Consistency if and only if it is a strictly increasing transformation of a generalized-Gini functional. In logarithmic coordinates, the corresponding result applies to strictly positive incomes with a fixed geometric mean and yields generalized log-Gini functionals. The two results are

absolute and relative versions of the same transformed-coordinate characterization.

Schur concavity orders the rank weights but does not itself produce the affine representation. Indeed, a continuous and strictly Schur-concave ranking may evaluate the same feasible displacement as improving at one baseline and worsening at another.

The characterization also extends beyond fixed arithmetic and geometric means. When the mean in the relevant coordinate system may vary, centering removes its change. In the absolute case, translation invariance in income space makes the ranking independent of the arithmetic mean. In the relative case, translation invariance in log-income space, equivalently scale invariance in income space, makes the ranking independent of the geometric mean. In both cases, these invariance conditions select a representation in which the coefficients attached to the ordered income ranks sum to zero.

These results identify the static rankings that admit an extension. The analysis also characterizes the dynamic indices that can extend each such ranking. Conditional on an admissible static ranking, a baseline-independent dynamic index may be any function of the relevant rank-based changes, provided that it classifies a transition as weakly improving exactly when the corresponding linear functional does. Static-Change Consistency determines how each transition is evaluated relative to no change, but it does not by itself determine how arbitrary pairs of transitions are ranked.

Dynamic Lorenz monotonicity adds precisely that further requirement. Among the characterized dynamic indices, those that are weakly monotone under Dynamic Lorenz dominance are those whose profile functionals preserve the cumulative-effect order. Full Dynamic Lorenz consistency additionally requires strict cumulative improvements to receive strictly larger values. Taking the difference between the corresponding generalized-Gini values yields a linear extension that is weakly monotone when the rank weights are decreasing and fully consistent when they are strictly decreasing.

The final results allow population sizes to vary, including transitions whose initial and final distributions have different sizes. Requiring the dynamic evaluation to be invariant when either distribution is replicated first implies static replication invariance. It also links, across population sizes, the coefficients attached to the ordered ranks in the fixed-population generalized-Gini representations. Initially, these rank coefficients agree only up to a positive factor. They can be rescaled consistently so that the coefficients attached to the finer ranks replacing an original rank sum exactly to the coefficient of that original rank. Their cumulative sums then define a concave function on rational population shares, and centered-quantile L^1 continuity extends this function continuously to $[0, 1]$.

The resulting theorems characterize both the static rankings that are coherent across population sizes and the dynamic indices that extend them. Among these indices, those that preserve unrestricted Dynamic Lorenz comparisons are exactly those that preserve the pointwise order of the normalized cumulative-effect curves.

Relation to earlier work. The use of linear rank weights is rooted in the generalized-Gini literature. Mehran (1976) studies linear inequality measures and the transfer principle. Weymark (1981) provides a static axiomatization of generalized-Gini absolute inequality orderings. His weak independence of income source requires an inequality comparison to remain unchanged when the same rank-ordered income component is added to both distributions. Together with the other static axioms, this condition yields linearity in ordered incomes, while the Pigou–Dalton transfer principle

orders the rank coefficients.

The present paper reaches the same generalized-Gini form from a different primitive. Rather than imposing independence directly on static comparisons, it asks whether the static ranking admits a baseline-independent dynamic extension satisfying Static-Change Consistency. Baseline independence yields linearity, while Schur concavity orders the rank coefficients. The paper also characterizes the complete class of dynamic indices that can extend each admissible static ranking.

Donaldson and Weymark (1980) study the single-parameter S-Gini subclass and the restrictions imposed across population sizes by Dalton's static population principle. Donaldson and Weymark (1983) extend the S-Ginis to continuum income distributions. The present variable-population analysis instead derives cross-population restrictions from independent replication of the initial and final distributions in a dynamic evaluation.

The closest comparison concerning change measurement is Bossert and Dutta (2019). Their framework begins with welfare-difference compatibility,

$$V(\mathbf{x}^0, \mathbf{x}^1) = W(\mathbf{x}^1) - W(\mathbf{x}^0),$$

and accordingly gives welfare differences cardinal significance. The present extension problem does not assume that the dynamic index is a difference of cardinal state values, nor does it impose positive linear homogeneity or weak independence of income source. It asks only that the dynamic evaluation be nonnegative exactly when the static change is weakly improving and that the evaluation depend on the relevant effect profile rather than separately on the baseline.

These requirements leave the cardinal form of the dynamic index less restricted and allow a larger class of dynamic indices. An admissible index must be nonnegative exactly when the corresponding generalized-Gini difference is nonnegative, but its magnitude need not be determined by that difference. The generalized-Gini difference is instead a linear member of the admissible class and provides the point of contact with cardinal welfare-change measurement. The present analysis also separates pure inequality change from aggregate movement by centering in identity or logarithmic coordinates, covers a relative log-income geometry, and extends the characterization to variable populations.

The affine representation is related to Candeal-Haro and Indurain-Eraso (1995), who obtain linear utility from global translation invariance on a topological vector space. The present domain need not be closed under arbitrary translations: equal displacements must generate the same comparison only when both occurrences are feasible. The argument is also analogous to affine representation on mixture spaces (Herstein and Milnor, 1953), although the independence condition here concerns common displacements rather than common mixtures.

The variable-population results are most closely related to static population principles for rank-dependent inequality and welfare orderings. Ebert (1988) uses Dalton's population principle to link fixed-population orderings across population sizes, while Sakamoto and Mori (2021) formulate variable-population rank-dependent welfare orderings through a common function of population shares. The present paper instead derives cross-population restrictions from independent replication of the initial and final distributions in a dynamic index. Because each fixed-population representation is initially determined only up to positive scale, the rank coefficients first agree only proportionally; their exact normalization and the continuity of the population-share weighting function must then be derived. Shorrocks (1983) provides a related use of replication to compare distributions of different sizes by representing them at a common population size.

The framework is anonymous and distributional. It compares ordered endpoint distributions and their rank-based changes, not the income paths of identified individuals. It therefore differs from mobility frameworks such as Chakravarty et al. (1985), which compare actual multiperiod income structures with hypothetical paths of complete immobility, and from the broader mobility literature surveyed by Fields and Ok (1999). Two transitions with the same ordered endpoint distributions are indistinguishable here even when their underlying individual trajectories differ.

Organization. Section 2 introduces the transformed-coordinate framework and proves the affine representation of baseline-independent orderings. Section 3 establishes the fixed- and variable-aggregate characterization theorems, characterizes the full class of admissible dynamic indices, and identifies those that preserve Dynamic Lorenz comparisons. Section 4 gives the absolute and relative applications. Section 5 distinguishes the ordinal extension problem from cardinal welfare-change measurement. Section 6 develops the variable-population characterizations. Sections 7 and 8 discuss the interpretation, scope, and principal conclusions.

2 Transformed-coordinate framework

2.1 Economic distributions and transformed-coordinate domains

An economic distribution records the ordered levels of an attribute, such as income, consumption, or wealth, across a population of fixed size. A static index evaluates the inequality of a single distribution, whereas a dynamic index evaluates the distributive change between an initial and a final distribution. The appropriate representation of such change depends on whether differences or ratios of economic levels are regarded as the relevant effects. We therefore conduct the analysis in transformed coordinates.

For each population size $n \geq 2$, let X_n denote an ordered economic domain. Its elements are admissible distributions

$$\mathbf{x} = (x_1, \ldots, x_n), \qquad x_1 \leq \cdots \leq x_n,$$

where coordinates represent ranks from the lowest to the highest level of the underlying economic attribute. The restrictions defining X_n depend on the application. For example, the absolute framework allows nonnegative incomes, whereas the relative framework requires strictly positive incomes.

An economic transition is an ordered pair

$$(\mathbf{x}, \mathbf{y}) \in X_n \times X_n,$$

where $\mathbf{x}$ is the initial distribution and $\mathbf{y}$ is the final distribution. The framework evaluates the distributive change between these two distributions. Because distributions are represented anonymously by their ordered coordinates, the transition records changes across ranks rather than individual income trajectories.

Let ψ be a strictly increasing continuous coordinate transformation, and define its coordinatewise extension by

$$\Psi_n(\mathbf{x}) = \big(\psi(x_1), \ldots, \psi(x_n)\big).$$

Because ψ is strictly increasing, Ψ_n preserves the rank ordering of the economic distribution. The transformed economic domain is $V_n^\psi = \Psi_n(X_n)$.

To formulate the representation results independently of a particular economic application, define the abstract ordered transformed-coordinate domain

$$V_n = \{\mathbf{v} \in \mathbb{R}^n : v_1 \leq \cdots \leq v_n\}.$$

Thus, $V_n^\psi \subseteq V_n$. The economic domain X_n specifies which distributions are admissible, while its image V_n^ψ expresses those distributions in the coordinates used to evaluate distributive effects. The larger domain V_n provides the ordered space in which the abstract geometric results are stated.

Given an economic transition $(\mathbf{x}, \mathbf{y})$, the coordinate transformation associates with its endpoints the ordered profiles

$$\Psi_n(\mathbf{x}) \qquad \text{and} \qquad \Psi_n(\mathbf{y}).$$

Their difference,

$$\boldsymbol{\tau}^\psi(\mathbf{x}, \mathbf{y}) = \Psi_n(\mathbf{y}) - \Psi_n(\mathbf{x}),$$

will be called the transformed-coordinate effect profile. Its ith component records the change between the initial and final levels occupying rank i, measured in the coordinate system selected by ψ.

The effect profile does not, by definition, contain all economically relevant information about the transition. In particular, it does not identify the initial distribution from which the effect is implemented. Whether the dynamic evaluation may nevertheless depend only on this profile is the substantive baseline-independence question studied below. Transition-Profile Independence will require two feasible transitions with the same transformed-coordinate effect profile to receive the same dynamic value. The representation theorems characterize the static inequality rankings for which such a dynamic evaluation is possible.

The choice of ψ determines how changes at each rank are measured. Under the identity transformation, $\psi_A(t) = t$, the transformed-coordinate effect profile consists of ordinary differences in economic levels and generates the absolute geometry. Under the logarithmic transformation,

$$\psi_R(t) = \log t,$$

it consists of log differences,

$$\log y_i - \log x_i = \log\left(\frac{y_i}{x_i}\right),$$

and therefore measures proportional changes, generating the relative geometry. The abstract representation results are proved for a general transformed-coordinate domain and are then applied to these identity and logarithmic specifications.

Anonymity and rank ordering. The framework evaluates distributions anonymously: only the economic levels at each rank matter, not the identities of the individuals who occupy those ranks. Accordingly, every economic distribution is represented by its increasing rearrangement,

$$x_1 \leq \cdots \leq x_n,$$

so that rank 1 is the poorest position and rank n is the richest. All economic distributions and transformed-coordinate profiles below are understood to be written in this ordered form. Cumulative

sums through rank k therefore refer to the poorest k positions. Because individual identities are not tracked across the initial and final distributions, the framework evaluates changes in ordered marginal distributions rather than individual mobility or reranking.

The Lorenz preorder and Schur concavity. On ordered transformed-coordinate profiles with the same total, we use the Lorenz preorder. For $\mathbf{v}, \mathbf{w} \in V_n$ satisfying

$$\sum_{i=1}^{n} v_i = \sum_{i=1}^{n} w_i,$$

write $\mathbf{v} \succcurlyeq_L \mathbf{w}$ if

$$\sum_{i=1}^{k} v_i \geq \sum_{i=1}^{k} w_i, \qquad \text{for every } k \in \{1, \ldots, n-1\}.$$

Because the coordinates are arranged increasingly, these inequalities compare the cumulative transformed-coordinate levels of the poorest k ranks. Thus $\mathbf{v} \succcurlyeq_L \mathbf{w}$ means that $\mathbf{v}$ Lorenz-dominates $\mathbf{w}$: among profiles with the same sum, $\mathbf{v}$ is at least as equal as $\mathbf{w}$.

A static index I^* is equality oriented if it respects the Lorenz preorder in the following direction:

$$\mathbf{v} \succcurlyeq_L \mathbf{w} \Rightarrow I^*(\mathbf{v}) \geq I^*(\mathbf{w}).$$

Equivalently, I^* is Schur concave. Thus, whenever $\mathbf{v} \succcurlyeq_L \mathbf{w}$, the more equal profile $\mathbf{v}$ receives a weakly larger value of I^*. Throughout the paper, larger values of I^* therefore represent greater equality, or equivalently, lower inequality.

The Lorenz preorder applies to static profiles whose transformed-coordinate levels have the same sum. The Dynamic Lorenz preorder introduced later applies instead to transitions and compares their cumulative distributive effects across ranks.

Centering and transformed-coordinate effects. For $\mathbf{v} \in V_n$, let

$$\bar{v} = \frac{1}{n} \sum_{i=1}^{n} v_i$$

denote its mean transformed-coordinate level, and define its centered profile by

$$C_n(\mathbf{v}) = \mathbf{v} - \bar{v}\mathbf{1}.$$

Centering removes the common transformed-coordinate level while retaining the profile's rank-dependent variation. By construction,

$$C_n(\mathbf{v}) \in H_0^n, \qquad H_0^n = \left\{ \mathbf{z} \in \mathbb{R}^n : \sum_{i=1}^{n} z_i = 0 \right\}.$$

For a transformed-coordinate transition $(\mathbf{v}, \mathbf{w})$, define its raw effect profile by

$$\boldsymbol{\tau}(\mathbf{v}, \mathbf{w}) = \mathbf{w} - \mathbf{v}.$$

Its ith component records the change at rank i in transformed coordinates. The profile may reflect both distributive variation across ranks and a uniform change affecting all ranks. To remove the latter component, define the centered effect profile by

$$\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}) = C_n(\mathbf{w}) - C_n(\mathbf{v}) = (\mathbf{w} - \mathbf{v}) - (\bar{w} - \bar{v})\mathbf{1}.$$

By construction,

$$\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}) \in H_0^n.$$

If $\bar{v} = \bar{w}$, no uniform adjustment is required, and therefore

$$\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}) = \boldsymbol{\tau}(\mathbf{v}, \mathbf{w}) = \mathbf{w} - \mathbf{v}.$$

2.2 Representing distributions on a common population-share scale

To compare ordered distributions with different population sizes, we represent each profile as a step function on the common interval $(0, 1]$. A point $p \in (0, 1]$ represents a cumulative population share. This construction places distributions with different numbers of observations on the same rank-based scale.

For $\mathbf{v} \in V_n$, define its ordered step profile by

$$v(p) = v_i \qquad \text{whenever } p \in \left(\frac{i-1}{n}, \frac{i}{n}\right], \quad i = 1, \ldots, n.$$

Thus v_1 is assigned to the poorest $1/n$ of the population, v_2 to the next $1/n$, and so forth, with v_n assigned to the richest $1/n$.

Because the analysis concerns inequality rather than the common level of the transformed coordinates, we center this step profile around its mean. Define

$$v^c(p) = v_i - \bar{v} \qquad \text{whenever } p \in \left(\frac{i-1}{n}, \frac{i}{n}\right], \quad i = 1, \ldots, n,$$

where

$$\bar{v} = \frac{1}{n} \sum_{i=1}^{n} v_i$$

is the mean transformed-coordinate level. The centered profile v^c records, at each population rank, the deviation from the profile's mean.

Profiles with different population sizes can now be compared on the same population-share interval. For $\mathbf{v} \in V_n$ and $\mathbf{w} \in V_m$, define

$$\|v^c - w^c\|_1 = \int_0^1 |v^c(p) - w^c(p)| \, dp.$$

This quantity measures the average absolute difference between the two centered rank profiles. It is well defined even when $n \neq m$ because both profiles are represented as functions on $(0, 1]$.

To formulate continuity across population sizes, define the cross-population transformed domain by

$$\mathcal{V}^{\psi} = \bigsqcup_{n \geq 2} V_n^{\psi}.$$

The symbol $\bigsqcup$ denotes a disjoint union. Thus an element of $\mathcal{V}^{\psi}$ consists of an admissible transformed-coordinate profile together with its population size. A cross-population static index is a single function $I^* : \mathcal{V}^{\psi} \to \mathbb{R}$, rather than a collection of unrelated fixed-population indices.

Definition 1 (Centered-Quantile L^1 Continuity)**.** A static index $I^* : \mathcal{V}^{\psi} \to \mathbb{R}$ satisfies centered-quantile L^1 continuity if, for every population size $n \geq 2$, every profile $\mathbf{v} \in V_n^{\psi}$, every sequence of population sizes $\{n_r\}_{r\geq 1}$, and every sequence of profiles $\{\mathbf{v}^r\}_{r\geq 1}$ satisfying

$$\mathbf{v}^r \in V_{n_r}^{\psi}, \qquad \text{for every } r \geq 1,$$

one has

$$\|(v^r)^c - v^c\|_1 \to 0 \Rightarrow I^*(\mathbf{v}^r) \to I^*(\mathbf{v}).$$

The population sizes n_r may vary with r and need not equal n.

The condition requires static evaluations to converge whenever the corresponding centered rank profiles become close in average absolute magnitude. Because convergence is imposed on centered profiles, the profiles may differ in their mean transformed-coordinate levels. What must converge is the pattern of deviations from the mean across population ranks. The approximating profiles may also have different and changing population sizes.

The domain $\mathcal{V}^{\psi}$ contains only transformed profiles generated by admissible finite economic distributions. The continuity condition does not require the static index to be defined on every function in $L^1(0, 1)$.

2.3 Dynamic extension principles

Fix a population size $n \geq 2$ and let $D \subseteq V_n$ be the ordered transformed-coordinate domain under consideration. Depending on the application, D may be a transformed economic domain V_n^{ψ}, a fixed-total subset of that domain, or a centered representation of it.

Let $I^* : D \to \mathbb{R}$ be a static index and $J^* : D \times D \to \mathbb{R}$ a dynamic index. The value $I^*(\mathbf{v})$ evaluates a profile $\mathbf{v} \in D$, whereas $J^*(\mathbf{v}, \mathbf{w})$ evaluates the transition from $\mathbf{v}$ to $\mathbf{w}$.

Definition 2 (Normalization and Static-Change Consistency)**.** The dynamic index J^* satisfies:

1. *Normalization* (N) if $J^*(\mathbf{v}, \mathbf{v}) = 0$ for every $\mathbf{v} \in D$.

2. *Static-Change Consistency* (SC) if

$$J^*(\mathbf{v}, \mathbf{w}) \geq 0 \Leftrightarrow I^*(\mathbf{w}) \geq I^*(\mathbf{v}), \qquad \text{for every } \mathbf{v}, \mathbf{w} \in D.$$

Normalization assigns zero to a transition with identical endpoints.

Static-Change Consistency has a natural induced-ranking interpretation. A dynamic evaluation can be used to compare two static profiles by evaluating the transition from $\mathbf{v}$ to $\mathbf{w}$ against the null transition at $\mathbf{v}$. Under the sign convention used here, $J^*(\mathbf{v}, \mathbf{w}) \geq 0$ means that the transition from $\mathbf{v}$ to $\mathbf{w}$ is weakly improving relative to no change. SC requires the static ranking induced by this comparison to coincide with the ranking represented by I^*:

$$J^*(\mathbf{v}, \mathbf{w}) \geq 0 \Leftrightarrow I^*(\mathbf{w}) \geq I^*(\mathbf{v}).$$

Thus, under the equality-oriented convention of the paper, the transition from $\mathbf{v}$ to $\mathbf{w}$ is weakly improving exactly when $\mathbf{w}$ is weakly more equal than $\mathbf{v}$ according to the maintained static index.

SC is a weak sign requirement. If $I^*(\mathbf{w}) = I^*(\mathbf{v})$, it implies $J^*(\mathbf{v}, \mathbf{w}) \geq 0$ and $J^*(\mathbf{w}, \mathbf{v}) \geq 0$, but it does not require either dynamic value to equal zero.

Definition 3 (Transition-Profile Independence)**.** Suppose that, for some $b \in \mathbb{R}$,

$$D \subseteq \left\{ \mathbf{v} \in V_n : \sum_{i=1}^{n} v_i = b \right\}.$$

The dynamic index J^* satisfies *Transition-Profile Independence* (TPI) if

$$\mathbf{w} - \mathbf{v} = \mathbf{w}' - \mathbf{v}' \Rightarrow J^*(\mathbf{v}, \mathbf{w}) = J^*(\mathbf{v}', \mathbf{w}'), \qquad \text{for all } \mathbf{v}, \mathbf{w}, \mathbf{v}', \mathbf{w}' \in D.$$

TPI requires transitions with the same transformed-coordinate effect profile to receive the same dynamic value. It is a baseline- independence condition: once the changes across ranks are specified, the initial profile carries no additional significance.

Definition 4 (Centered Transition-Profile Independence)**.** Suppose that D permits variation in the mean transformed-coordinate level. The dynamic index J^* satisfies *Centered Transition-Profile Independence* (CTPI) if

$$\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}) = \widetilde{\boldsymbol{\tau}}(\mathbf{v}', \mathbf{w}') \Rightarrow J^*(\mathbf{v}, \mathbf{w}) = J^*(\mathbf{v}', \mathbf{w}'), \qquad \text{for all } \mathbf{v}, \mathbf{w}, \mathbf{v}', \mathbf{w}' \in D.$$

CTPI requires transitions with the same centered effect profile to receive the same dynamic value. Centering removes the uniform component of transformed-coordinate change. The resulting evaluation therefore depends neither on the baseline profile nor on the change in the mean transformed-coordinate level.

On a fixed-total domain, the endpoints of every transition have the same transformed-coordinate mean, so $\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}) = \mathbf{w} - \mathbf{v}$. TPI is therefore the fixed-total counterpart of CTPI.

The fixed-total characterization below uses N, SC, and TPI. When the mean transformed-coordinate level may vary, TPI is replaced by CTPI. Additional replication and profile-independence conditions are introduced when population size is allowed to vary.

2.4 Affine representation of baseline-independent orderings

The main characterization relies on an implication of baseline independence. Suppose that two feasible transitions generate the same change in transformed coordinates. Baseline independence requires that this change be evaluated in the same direction regardless of the initial profile from which it occurs. Thus, whether a feasible displacement is improving, neutral, or worsening depends on the displacement itself rather than on its location within the domain.

The result below shows that baseline independence imposes considerable structure on the static ranking and, consequently, on the dynamic indices that can extend it. If the evaluation of a feasible displacement is independent of the baseline from which it occurs, the same criterion must classify that displacement as improving, neutral, or worsening wherever it can be implemented. Continuity

and consistency of the underlying ranking ensure that these classifications fit together coherently. As a result, all feasible displacements can be evaluated by a single linear score:

$$\mathbf{w} \succeq \mathbf{v} \Leftrightarrow L(\mathbf{w} - \mathbf{v}) \geq 0.$$

Thus, the comparison between two profiles depends on the sign of one linear function of the change between them.

The main technical issue is that the economic domains considered in the paper may be restricted by rank ordering, nonnegativity, or aggregate conditions. A displacement that is feasible from one profile therefore need not be feasible from every other profile. Standard linear representation results typically avoid this issue by assuming that the ranking is defined on an entire vector space and that every translation is admissible; see Candeal-Haro and Indurain-Eraso (1995). The present framework requires less: equal displacements must receive the same comparison only when both occurrences are feasible within the maintained domain. The representation is established directly on that domain, without first extending the ranking to a larger unrestricted space.

The argument is also related to affine utility representation on mixture spaces; see Herstein and Milnor (1953). The analogy lies in the use of an independence requirement to obtain an affine representation. The primitives differ, however: mixture-space independence concerns comparisons preserved under common mixtures, whereas the present condition concerns comparisons preserved when the same feasible displacement is implemented from different baselines.

The proof proceeds in two steps. We first show that the weakly improving displacements form a coherent set of directions. We then show that this set can be represented by the nonnegative values of a linear functional. Finally, continuity extends the representation from comparisons between interior profiles to comparisons involving boundary profiles.

We use the following geometric notation. The domain D may lie in a proper affine subspace of the surrounding Euclidean space. An affine space is a translated vector space and therefore need not contain the origin. The relative interior of D, denoted by $\mathrm{ri}(D)$, is its interior within the smallest affine space containing it. A point in $\mathrm{ri}(D)$ permits sufficiently small movements in every direction parallel to that affine space, even when D has empty interior in the surrounding Euclidean space.

The set

$$D - D = \{\mathbf{w} - \mathbf{v} : \mathbf{v}, \mathbf{w} \in D\}$$

contains the displacements generated by pairs of profiles in D. Let $V = \mathrm{span}(D - D)$ denote the vector space generated by those displacements. Thus V contains every direction that can be formed from feasible changes within D.

A subset $P \subseteq V$ is a convex cone if

$$\mathbf{y}, \mathbf{z} \in P,\ \alpha, \beta \geq 0 \Rightarrow \alpha\mathbf{y} + \beta\mathbf{z} \in P.$$

Thus a convex cone contains every nonnegative linear combination of its elements. For any set $P \subseteq V$, write

$$-P = \{-\mathbf{z} : \mathbf{z} \in P\}$$

for the set obtained by reversing the direction of every vector in P.

In the lemma below, P will consist of the displacement vectors $\mathbf{z} \in V$ for which a sufficiently small movement from a reference profile $\mathbf{a}$ to $\mathbf{a} + \varepsilon\mathbf{z}$ is weakly improving.

Lemma 1 (Directional-cone representation)**.** *Let D be a convex subset of a finite-dimensional affine space with nonempty relative interior* $\mathrm{ri}(D)$*. Let $\succcurlyeq$ be a complete, transitive, and nonconstant preorder on D whose upper and lower contour sets are closed in the relative topology of D. Suppose that equal feasible displacements generate the same comparison: whenever $\mathbf{w} - \mathbf{v} = \mathbf{w}' - \mathbf{v}'$ and $\mathbf{v}, \mathbf{w}, \mathbf{v}', \mathbf{w}' \in D$, one has*

$$\mathbf{w} \succcurlyeq \mathbf{v} \Leftrightarrow \mathbf{w}' \succcurlyeq \mathbf{v}'.$$

Fix a reference point $\mathbf{a} \in \mathrm{ri}(D)$. Define $P \subseteq V$ by declaring that $\mathbf{z} \in P$ if there exists $\delta > 0$ such that

$$\mathbf{a} + \varepsilon\mathbf{z} \in D \quad \textit{and} \quad \mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$$

for every $\varepsilon \in (0, \delta)$. Then:

1. *The definition of P is independent of the relative-interior reference point used in its construction.*
2. *P is a closed convex cone.*
3. *$P \cup (-P) = V$ and $P \neq V$. Thus, for every direction $\mathbf{z} \in V$, either $\mathbf{z}$ or its opposite $-\mathbf{z}$ is weakly improving, but not every direction is weakly improving.*
4. *For every $\mathbf{z} \in V$ and every $\varepsilon > 0$ such that $\mathbf{a} + \varepsilon\mathbf{z} \in D$, one has*

$$\mathbf{z} \in P \Leftrightarrow \mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}.$$

The lemma gives a baseline-independent classification of displacement directions. A direction belongs to P precisely when a feasible movement in that direction is weakly improving. This classification is independent of the relative-interior reference profile used to construct it and does not depend on the scale at which the direction is implemented. The next lemma shows that a cone with these properties can be represented by the nonnegative values of a linear functional.

Proof of Lemma 1. Write $\succ$ for the strict part of $\succcurlyeq$. Feasible-displacement independence transfers strict comparisons as well. Indeed, if $\mathbf{w} - \mathbf{v} = \mathbf{w}' - \mathbf{v}'$ and $\mathbf{w} \succ \mathbf{v}$, then $\mathbf{w} \succcurlyeq \mathbf{v}$ and not $\mathbf{v} \succcurlyeq \mathbf{w}$. Applying feasible-displacement independence to the displayed displacement and to its negative gives $\mathbf{w}' \succcurlyeq \mathbf{v}'$ and not $\mathbf{v}' \succcurlyeq \mathbf{w}'$, hence $\mathbf{w}' \succ \mathbf{v}'$.

Independence of the reference profile. Let $\mathbf{a}, \mathbf{a}' \in \mathrm{ri}(D)$ and let $\mathbf{z} \in V$. Because both reference profiles belong to the relative interior of D, for every sufficiently small $\varepsilon > 0$ the four points

$$\mathbf{a}, \quad \mathbf{a} + \varepsilon\mathbf{z}, \quad \mathbf{a}', \quad \mathbf{a}' + \varepsilon\mathbf{z}$$

belong to D. The two transitions generate the same displacement:

$$(\mathbf{a} + \varepsilon\mathbf{z}) - \mathbf{a} = \varepsilon\mathbf{z} = (\mathbf{a}' + \varepsilon\mathbf{z}) - \mathbf{a}'.$$

Consequently, feasible-displacement independence gives

$$\mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a} \Leftrightarrow \mathbf{a}' + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}'.$$

Hence a direction is locally weakly improving from $\mathbf{a}$ if and only if it is locally weakly improving from $\mathbf{a}'$. Therefore, the set P is independent of the relative-interior reference profile used in its definition.

Independence from scale and proof of item 4. We now show that the classification of a direction does not depend on the feasible scale at which it is considered. More precisely, we prove that

$$\mathbf{z} \in P \Leftrightarrow \mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$$

for every $\varepsilon > 0$ such that $\mathbf{a} + \varepsilon\mathbf{z} \in D$. This passage proves item 4. It also establishes positive homogeneity: if $\mathbf{z} \in P$, then $\lambda\mathbf{z} \in P$ for every $\lambda \geq 0$. Positive homogeneity is one component of the assertion in item 2 that P is a convex cone. The other component, closure under addition, and the topological closedness of P are proved below.

Suppose first that $\mathbf{a} + \varepsilon_0\mathbf{z} \succcurlyeq \mathbf{a}$ for some $\varepsilon_0 > 0$ such that $\mathbf{a} + \varepsilon_0\mathbf{z} \in D$. Because D is convex, the entire segment from $\mathbf{a}$ to $\mathbf{a} + \varepsilon_0\mathbf{z}$ belongs to D. For an integer $m \geq 1$, define the equally spaced points

$$\mathbf{a}_j = \mathbf{a} + \frac{j\varepsilon_0}{m}\mathbf{z}, \qquad j \in \{0, \ldots, m\}.$$

We first show that the initial increment is not strictly worsening. Suppose, to the contrary, that $\mathbf{a}_0 \succ \mathbf{a}_1$. Every reversed adjacent pair generates the same displacement:

$$\mathbf{a}_{j-1} - \mathbf{a}_j = -\frac{\varepsilon_0}{m}\mathbf{z} = \mathbf{a}_0 - \mathbf{a}_1, \qquad j \in \{1, \ldots, m\}.$$

Therefore, feasible-displacement independence, applied to each adjacent pair and to the corresponding reversed comparison, gives $\mathbf{a}_{j-1} \succ \mathbf{a}_j$ for every $j \in \{1, \ldots, m\}$. Transitivity of the strict part of the preorder then implies $\mathbf{a}_0 \succ \mathbf{a}_m$. But

$$\mathbf{a}_m = \mathbf{a} + \varepsilon_0\mathbf{z} \succcurlyeq \mathbf{a} = \mathbf{a}_0,$$

which is a contradiction. Hence $\mathbf{a}_1 \succcurlyeq \mathbf{a}_0$.

Every forward adjacent pair also generates the same displacement:

$$\mathbf{a}_j - \mathbf{a}_{j-1} = \frac{\varepsilon_0}{m}\mathbf{z} = \mathbf{a}_1 - \mathbf{a}_0.$$

Consequently, feasible-displacement independence gives $\mathbf{a}_j \succcurlyeq \mathbf{a}_{j-1}$ for every $j \in \{1, \ldots, m\}$. By transitivity, $\mathbf{a}_k \succcurlyeq \mathbf{a}_0$ for every $k \in \{1, \ldots, m\}$, or equivalently,

$$\mathbf{a} + \frac{k\varepsilon_0}{m}\mathbf{z} \succcurlyeq \mathbf{a}, \qquad k \in \{1, \ldots, m\}.$$

The same argument applies after any further subdivision. Hence the comparison holds at every rational subdivision scale in $[0, \varepsilon_0]$. Approximating any real $\varepsilon \in [0, \varepsilon_0]$ by such scales and using the closedness of the upper contour set of $\mathbf{a}$ gives $\mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$ whenever the displaced point is feasible.

Conversely, suppose that $\mathbf{z} \in P$. By the definition of P, there exists $\delta_0 > 0$ such that $\mathbf{a} + \delta\mathbf{z} \succcurlyeq \mathbf{a}$ for every $\delta \in (0, \delta_0)$. Let $\varepsilon > 0$ satisfy $\mathbf{a} + \varepsilon\mathbf{z} \in D$. Choose m sufficiently large that $\frac{\varepsilon}{m} < \delta_0$. Define

$$\mathbf{b}_j = \mathbf{a} + \frac{j\varepsilon}{m}\mathbf{z}, \qquad j \in \{0, \ldots, m\}.$$

Convexity of D makes every $\mathbf{b}_j$ feasible. The definition of P gives $\mathbf{b}_1 \succcurlyeq \mathbf{b}_0$. Since every adjacent pair has the same displacement, feasible-displacement independence yields

$$\mathbf{b}_j \succcurlyeq \mathbf{b}_{j-1}, \qquad j \in \{1, \ldots, m\}.$$

Therefore, transitivity gives

$$\mathbf{a} + \varepsilon\mathbf{z} = \mathbf{b}_m \succcurlyeq \mathbf{b}_0 = \mathbf{a}.$$

We have proved that

$$\mathbf{z} \in P \Leftrightarrow \mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$$

at every feasible positive scale ε, which establishes item 4. It also follows that

$$\mathbf{z} \in P, \quad \lambda \geq 0 \Rightarrow \lambda\mathbf{z} \in P.$$

The remaining part of item 2 is to show that P is closed under addition and is topologically closed.

Closure under addition and convexity. We next prove the remaining algebraic part of item 2. Let $\mathbf{y}, \mathbf{z} \in P$. The reference point $\mathbf{a} \in \mathrm{ri}(D)$ remains fixed as in the statement of the lemma. Choose $\varepsilon > 0$ sufficiently small that

$$\mathbf{a}, \qquad \mathbf{a} + \varepsilon\mathbf{y}, \qquad \mathbf{a} + \varepsilon(\mathbf{y} + \mathbf{z})$$

belong to $\mathrm{ri}(D)$.

Since $\mathbf{y} \in P$, item 4 gives $\mathbf{a} + \varepsilon\mathbf{y} \succcurlyeq \mathbf{a}$. The point $\mathbf{a} + \varepsilon\mathbf{y}$ also belongs to $\mathrm{ri}(D)$. Since the definition of P is independent of the relative-interior reference profile and $\mathbf{z} \in P$, movement by $\varepsilon\mathbf{z}$ from this new reference profile is weakly improving:

$$\mathbf{a} + \varepsilon(\mathbf{y} + \mathbf{z}) \succcurlyeq \mathbf{a} + \varepsilon\mathbf{y}.$$

Transitivity gives

$$\mathbf{a} + \varepsilon(\mathbf{y} + \mathbf{z}) \succcurlyeq \mathbf{a}.$$

By item 4, $\mathbf{y} + \mathbf{z} \in P$.

Thus P is closed under addition. Together with the positive homogeneity established above, this shows that P is a convex cone. The topological closedness of P, which is the remaining part of item 2, is proved below.

Completeness and closedness. For $\mathbf{z} \in V$, choose $\varepsilon > 0$ sufficiently small that $\mathbf{a} \pm \varepsilon\mathbf{z} \in D$. Completeness applied to $\mathbf{a} + \varepsilon\mathbf{z}$ and $\mathbf{a}$ gives either $\mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$, in which case $\mathbf{z} \in P$, or $\mathbf{a} \succcurlyeq \mathbf{a} + \varepsilon\mathbf{z}$. In the latter case, feasible-displacement independence applied to

$$\mathbf{a} + \varepsilon\mathbf{z}, \quad \mathbf{a}, \quad \mathbf{a}, \quad \mathbf{a} - \varepsilon\mathbf{z}$$

gives $\mathbf{a} - \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$, hence $-\mathbf{z} \in P$. Therefore $P \cup (-P) = V$.

Let $\mathbf{z}_r \in P$ and $\mathbf{z}_r \to \mathbf{z}$. Choose $\varepsilon > 0$ such that $\mathbf{a} + \varepsilon\mathbf{z} \in \mathrm{ri}(D)$. For all sufficiently large r, $\mathbf{a} + \varepsilon\mathbf{z}_r \in D$, and item 4 gives

$$\mathbf{a} + \varepsilon\mathbf{z}_r \succcurlyeq \mathbf{a}.$$

Closedness of the upper contour set of $\mathbf{a}$ yields $\mathbf{a} + \varepsilon\mathbf{z} \succcurlyeq \mathbf{a}$, hence $\mathbf{z} \in P$.

Properness. It remains to prove the second assertion in item 3, namely that $P \neq V$. Suppose, to the contrary, that $P = V$. Let $\mathbf{v}, \mathbf{w} \in \mathrm{ri}(D)$ and put $\mathbf{d} = \mathbf{w} - \mathbf{v}$. Since $P = V$, both

$$\mathbf{d} \in P \qquad \text{and} \qquad -\mathbf{d} \in P.$$

Choose an integer m sufficiently large that the equally spaced points

$$\mathbf{v}_j = \mathbf{v} + \frac{j}{m}\mathbf{d}, \qquad j \in \{0, \ldots, m\},$$

belong to $\mathrm{ri}(D)$. For each $j \in \{1, \dots, m\}$, the displacement from $\mathbf{v}_{j-1}$ to $\mathbf{v}_j$ is

$$\mathbf{v}_j - \mathbf{v}_{j-1} = \frac{1}{m}\mathbf{d}.$$

Because $\mathbf{d} \in P$, positive homogeneity gives $\frac{1}{m}\mathbf{d} \in P$. Therefore, independence of the reference profile and item 4 imply

$$\mathbf{v}_j \succcurlyeq \mathbf{v}_{j-1}, \qquad j \in \{1, \dots, m\}.$$

Transitivity yields

$$\mathbf{w} = \mathbf{v}_m \succcurlyeq \mathbf{v}_0 = \mathbf{v}.$$

Similarly, the reverse displacement is

$$\mathbf{v}_{j-1} - \mathbf{v}_j = -\frac{1}{m}\mathbf{d}.$$

Because $-\mathbf{d} \in P$, positive homogeneity gives $-\frac{1}{m}\mathbf{d} \in P$. Therefore, independence of the reference profile and item 4 imply

$$\mathbf{v}_{j-1} \succcurlyeq \mathbf{v}_j, \qquad j \in \{1, \dots, m\}.$$

Transitivity yields $\mathbf{v} = \mathbf{v}_0 \succcurlyeq \mathbf{v}_m = \mathbf{w}$.

Hence $\mathbf{v} \sim \mathbf{w}$. Thus all relative-interior profiles are indifferent.

For arbitrary $\mathbf{v}, \mathbf{w} \in D$, fix $\mathbf{a} \in \mathrm{ri}(D)$ and define

$$\mathbf{v}_t = (1-t)\mathbf{v} + t\mathbf{a}, \qquad \mathbf{w}_t = (1-t)\mathbf{w} + t\mathbf{a}, \qquad t \in (0, 1).$$

Then $\mathbf{v}_t, \mathbf{w}_t \in \mathrm{ri}(D)$. Since all relative-interior profiles are indifferent, $\mathbf{v}_t \sim \mathbf{a}$ and $\mathbf{w}_t \sim \mathbf{a}$ for every $t \in (0, 1)$. As $t \downarrow 0$, $\mathbf{v}_t \to \mathbf{v}$ and $\mathbf{w}_t \to \mathbf{w}$. Closedness of the upper and lower contour sets of $\mathbf{a}$ therefore gives $\mathbf{v} \sim \mathbf{a}$ and $\mathbf{w} \sim \mathbf{a}$. By transitivity, $\mathbf{v} \sim \mathbf{w}$. Therefore, the preorder would be constant on D, contrary to nonconstancy. Therefore, $P \neq V$. ■

Lemma 1 gives a baseline-independent classification of displacement directions: a direction belongs to P precisely when a feasible movement in that direction is weakly improving. This classification is independent of the relative-interior reference profile and of the feasible scale at which the direction is implemented. It remains to show that membership in P can be determined by the sign of a linear functional.

The next lemma converts the directional classification obtained above into a numerical one. Its proof removes the directions that are classified as weakly improving in both orientations. After those neutral directions are identified, the remaining comparison is one-dimensional and can therefore be represented by the sign of a linear functional.

For clarity, let N be a linear subspace of V. For each $\mathbf{z} \in V$, define

$$\mathbf{z} + N = \{\mathbf{z} + \mathbf{y} : \mathbf{y} \in N\}.$$

This is the set of vectors obtained from $\mathbf{z}$ by adding elements of N.

Two vectors $\mathbf{z}, \mathbf{z}' \in V$ generate the same such set precisely when their difference belongs to N:

$$\mathbf{z} + N = \mathbf{z}' + N \Leftrightarrow \mathbf{z} - \mathbf{z}' \in N.$$

Indeed, if $\mathbf{z} - \mathbf{z}' = \mathbf{y}_0 \in N$, then, for every $\mathbf{y} \in N$,

$$\mathbf{z} + \mathbf{y} = \mathbf{z}' + (\mathbf{y}_0 + \mathbf{y}),$$

and $\mathbf{y}_0 + \mathbf{y} \in N$. Hence $\mathbf{z} + N \subseteq \mathbf{z}' + N$. Reversing the roles of $\mathbf{z}$ and $\mathbf{z}'$ gives the opposite inclusion. Conversely, if the two sets coincide, then $\mathbf{z} \in \mathbf{z}' + N$, so $\mathbf{z} = \mathbf{z}' + \mathbf{y}$ for some $\mathbf{y} \in N$, and therefore $\mathbf{z} - \mathbf{z}' \in N$.

The quotient space V/N is the collection of all these sets:

$$V/N = \{\mathbf{z} + N : \mathbf{z} \in V\}.$$

Passing from V to V/N identifies vectors that differ only by an element of N. The canonical quotient map is

$$\pi : V \to V/N, \qquad \pi(\mathbf{z}) = \mathbf{z} + N.$$

For a set $P \subseteq V$, write

$$P/N = \pi(P) = \{\mathbf{z} + N : \mathbf{z} \in P\}$$

for its image in the quotient space.

A cone C is *pointed* if

$$C \cap (-C) = \{0\}.$$

Thus, no nonzero vector and its opposite both belong to C. The condition

$$C \cup (-C) = X$$

means that, for every $\mathbf{x} \in X$, either $\mathbf{x} \in C$ or $-\mathbf{x} \in C$. Together, these conditions imply that every nonzero vector belongs to exactly one of C and $-C$.

Lemma 2 (Half-space representation)**.** *Let V be a finite-dimensional vector space and let $P \subseteq V$ be a closed convex cone satisfying*

$$P \cup (-P) = V \qquad \text{and} \qquad P \neq V.$$

Then there exists a nonzero linear functional $L : V \to \mathbb{R}$ such that

$$P = \{\mathbf{z} \in V : L(\mathbf{z}) \geq 0\}.$$

Proof. Let $N = P \cap (-P)$. Thus N contains the directions that belong to P together with their opposites. These are the directions treated as neutral by the cone.

We first verify that N is a linear subspace of V. Since P and $-P$ are convex cones, their intersection is closed under addition and multiplication by nonnegative scalars. Moreover, if $\mathbf{z} \in N$, then both $\mathbf{z}$ and $-\mathbf{z}$ belong to P. Hence $-\mathbf{z} \in N$ as well. Therefore N is also closed under multiplication by negative scalars and is a linear subspace.

We now remove these neutral directions by passing to the quotient space $X = V/N$. Let $\pi : V \to X$ be the quotient map, and let $C = \pi(P) = P/N$. Two vectors have the same image under π precisely when their difference belongs to N. Thus X treats directions that differ only by a neutral direction as equivalent.

Because $N \subseteq P$ and P is closed under addition, $P + N = P$. Consequently, if one vector belongs to P, then every vector that differs from it by an element of N also belongs to P. Equivalently,

P is the full inverse image of C: $P = \pi^{-1}(C)$. Since $X = V/N$ carries the quotient topology and $P = \pi^{-1}(C)$, the closedness of P implies that C is closed in X. Linearity of π and convexity of P imply that C is a convex cone.

We next show that, after passing to the quotient space, no nonzero direction belongs to the image of P in both orientations:

$$C \cap (-C) = \{0\}.$$

Let $\mathbf{x} \in C \cap (-C)$. Since $\mathbf{x} \in C = \pi(P)$, there exists $\mathbf{p} \in P$ such that $\mathbf{x} = \pi(\mathbf{p})$. Since $-\mathbf{x} \in C$, there also exists $\mathbf{q} \in P$ such that $-\mathbf{x} = \pi(\mathbf{q})$. Therefore,

$$\pi(\mathbf{p} + \mathbf{q}) = \pi(\mathbf{p}) + \pi(\mathbf{q}) = \mathbf{x} - \mathbf{x} = 0.$$

By the definition of the quotient map, $\pi(\mathbf{p} + \mathbf{q}) = 0$ means that $\mathbf{p} + \mathbf{q} \in N$.

Since $N = P \cap (-P)$, membership in N implies that $-(\mathbf{p} + \mathbf{q}) \in P$. Because $\mathbf{q} \in P$ and P is closed under addition,

$$-\mathbf{p} = -(\mathbf{p} + \mathbf{q}) + \mathbf{q} \in P.$$

We already have $\mathbf{p} \in P$, so $\mathbf{p} \in P \cap (-P) = N$. Hence $\mathbf{x} = \pi(\mathbf{p}) = 0$.

Thus the only element belonging to both C and $-C$ is the zero element: $C \cap (-C) = \{0\}$.

The condition $P \cup (-P) = V$ also passes to the quotient:

$$C \cup (-C) = X.$$

Thus every nonzero vector of X belongs to exactly one of C and $-C$. Moreover, X is not the zero space. Otherwise, $V = N$ would imply $V \subseteq P$, contradicting $P \neq V$.

We next show that X is one dimensional. Suppose, to the contrary, that $\dim X \geq 2$. Since $C \cup (-C) = X$, the vectors in C span X. Therefore, we can choose two linearly independent vectors $\mathbf{u}, \mathbf{v} \in C$.

Consider the line

$$\mathcal{L} = \{\alpha\mathbf{u} + (1 - \alpha)(-\mathbf{v}) : \alpha \in \mathbb{R}\}.$$

This line contains $\mathbf{u} \in C$ when $\alpha = 1$ and $-\mathbf{v} \in -C$ when $\alpha = 0$. Since every point of X belongs to C or $-C$, the line is covered by $\mathcal{L} \cap C$ and $\mathcal{L} \cap (-C)$. Both sets are nonempty and closed relative to $\mathcal{L}$.

The line $\mathcal{L}$ is connected. It therefore cannot be written as the union of two disjoint nonempty relatively closed sets. Hence $\mathcal{L} \cap C$ and $\mathcal{L} \cap (-C)$ must intersect. Their intersection is contained in

$$C \cap (-C) = \{0\},$$

so this would require $0 \in \mathcal{L}$.

But $0 \in \mathcal{L}$ would mean that $\alpha\mathbf{u} = (1-\alpha)\mathbf{v}$ for some $\alpha \in \mathbb{R}$, contradicting the linear independence of $\mathbf{u}$ and $\mathbf{v}$. We conclude that $\dim X = 1$.

Because X is a nonzero one-dimensional vector space, its elements can be assigned a single linear coordinate. Choose a nonzero linear functional $\ell : X \to \mathbb{R}$. Replacing ℓ by $-\ell$ if necessary, we may choose its orientation so that the vectors in C receive nonnegative values:

$$C = \{\mathbf{x} \in X : \ell(\mathbf{x}) \geq 0\}.$$

Indeed, C contains exactly one vector from each pair $\{\mathbf{x}, -\mathbf{x}\}$ of opposite nonzero vectors in X.

Finally, define $L = \ell \circ \pi$. This is a linear functional on V. It is nonzero because π maps V onto the nonzero space X and ℓ is nonzero. For every $\mathbf{z} \in V$,

$$\mathbf{z} \in P \Leftrightarrow \pi(\mathbf{z}) \in C \Leftrightarrow \ell(\pi(\mathbf{z})) \geq 0 \Leftrightarrow L(\mathbf{z}) \geq 0.$$

Therefore, $P = \{\mathbf{z} \in V : L(\mathbf{z}) \geq 0\}$. ∎

The two preceding lemmas complete the geometric part of the argument. Lemma 1 shows that baseline-independent comparisons classify displacement directions through the cone P. Lemma 2 shows that this classification is equivalent to the sign of a single linear functional. It remains to translate that directional representation back into comparisons between profiles in D, including profiles on the boundary of the domain.

Theorem 1 (Affine representation of baseline-independent orderings)**.** *Under the hypotheses of Lemma 1, there exists a nonzero linear functional*

$$L : \operatorname{span}(D - D) \to \mathbb{R}$$

such that

$$\mathbf{w} \succcurlyeq \mathbf{v} \Leftrightarrow L(\mathbf{w} - \mathbf{v}) \geq 0, \qquad \text{for all } \mathbf{v}, \mathbf{w} \in D.$$

Proof. Let

$$V = \operatorname{span}(D - D).$$

By Lemma 1, the weakly improving displacement directions form a set $P \subseteq V$ with the following properties:

1. the definition of P is independent of the relative-interior reference profile;
2. a direction belongs to P if and only if movement in that direction is weakly improving at every feasible positive scale;
3. for every $\mathbf{z} \in V$, either $\mathbf{z} \in P$ or $-\mathbf{z} \in P$.

By Lemma 2, there is a nonzero linear functional $L : V \to \mathbb{R}$ such that

$$P = \{\mathbf{z} \in V : L(\mathbf{z}) \geq 0\}.$$

Thus the two lemmas reduce the desired representation to showing that, for any $\mathbf{v}, \mathbf{w} \in D$, the comparison $\mathbf{w} \succcurlyeq \mathbf{v}$ is equivalent to the membership condition $\mathbf{w} - \mathbf{v} \in P$.

Fix $\mathbf{v}, \mathbf{w} \in D$ and let $\mathbf{d} = \mathbf{w} - \mathbf{v}$. Also fix a reference profile $\mathbf{a} \in \operatorname{ri}(D)$. Since $\mathbf{d} \in V$ and $\mathbf{a} \in \operatorname{ri}(D)$, there exists an integer m large enough that $\mathbf{a} + \frac{\mathbf{d}}{m} \in D$ and $\mathbf{a} - \frac{\mathbf{d}}{m} \in D$. By convexity of D, the equally spaced points

$$\mathbf{v}_j = \mathbf{v} + \frac{j}{m}\mathbf{d}, \qquad j \in \{0, \dots, m\},$$

also belong to D.

Suppose first that $L(\mathbf{d}) \geq 0$. Then $\mathbf{d} \in P$, and positive homogeneity of P gives $\mathbf{d}/m \in P$. Lemma 1(4) gives $\mathbf{a} + \mathbf{d}/m \succcurlyeq \mathbf{a}$.

Every adjacent pair in the partition has the same displacement:

$$\mathbf{v}_j - \mathbf{v}_{j-1} = \frac{1}{m}\mathbf{d} = \left(\mathbf{a} + \frac{1}{m}\mathbf{d}\right) - \mathbf{a}.$$

Therefore, feasible-displacement independence gives $\mathbf{v}_j \succcurlyeq \mathbf{v}_{j-1}$ for every $j \in \{1, \dots, m\}$. Transitivity then yields $\mathbf{w} = \mathbf{v}_m \succcurlyeq \mathbf{v}_0 = \mathbf{v}$.

Conversely, suppose that $\mathbf{w} \succcurlyeq \mathbf{v}$. We show that $L(\mathbf{d}) \geq 0$. Assume, to the contrary, that $L(\mathbf{d}) < 0$. Then $-\mathbf{d} \in P$, whereas $\mathbf{d} \notin P$. Positive homogeneity gives $-\mathbf{d}/m \in P$. Since $\mathbf{a} - \mathbf{d}/m \in D$ for every sufficiently large m, Lemma 1(4) gives $\mathbf{a} - \mathbf{d}/m \succcurlyeq \mathbf{a}$.

For every $j \in \{1, \dots, m\}$, the reverse adjacent displacement is $\mathbf{v}_{j-1} - \mathbf{v}_j = -\mathbf{d}/m$. Consequently, feasible-displacement independence gives $\mathbf{v}_{j-1} \succcurlyeq \mathbf{v}_j$.

The forward comparison cannot also hold. Indeed, if $\mathbf{v}_j \succcurlyeq \mathbf{v}_{j-1}$ for some j, then the equality

$$\mathbf{v}_j - \mathbf{v}_{j-1} = \frac{1}{m}\mathbf{d} = \left(\mathbf{a} + \frac{1}{m}\mathbf{d}\right) - \mathbf{a}$$

and feasible-displacement independence would give $\mathbf{a} + \mathbf{d}/m \succcurlyeq \mathbf{a}$. By Lemma 1(4), this would imply $\mathbf{d} \in P$, contrary to $\mathbf{d} \notin P$.

Hence $\mathbf{v}_{j-1} \succ \mathbf{v}_j$ for every $j \in \{1, \dots, m\}$. Transitivity of the strict part of the preorder gives $\mathbf{v} = \mathbf{v}_0 \succ \mathbf{v}_m = \mathbf{w}$, contradicting $\mathbf{w} \succcurlyeq \mathbf{v}$. Therefore $L(\mathbf{d}) \geq 0$.

Since $\mathbf{d} = \mathbf{w} - \mathbf{v}$, we have proved

$$\mathbf{w} \succcurlyeq \mathbf{v} \Leftrightarrow L(\mathbf{w} - \mathbf{v}) \geq 0$$

for every $\mathbf{v}, \mathbf{w} \in D$. ■

3 Fixed-population characterizations

3.1 Why Schur concavity is not enough

Schur concavity and baseline independence impose different restrictions on a static ranking. Schur concavity governs comparisons ordered by the Lorenz preorder: when two profiles have the same sum and one is more equal than the other in the Lorenz sense, the more equal profile must receive a weakly larger static value. It does not, however, determine the evaluation of every feasible change between profiles.

Baseline independence adds a different requirement. Together with Static-Change Consistency, it requires a given displacement to be weakly improving either at every baseline where it is feasible or at none of them. Thus a displacement that is weakly improving from one initial profile cannot be worsening from another if the two transitions generate the same effect profile. As shown in Section 2.4, this additional requirement is what forces all feasible displacements to be evaluated by a common linear functional.

The following example demonstrates why Schur concavity alone does not provide this structure. It presents a familiar continuous, strictly Schur-concave ranking and a single feasible displacement whose effect on the static index changes sign when it is implemented from different initial profiles.

Consider the equality-oriented index

$$I(\mathbf{x}) = -\sum_{i=1}^{3}(x_i - \mu_{\mathbf{x}})^2.$$

On a fixed-mean domain, this is the negative of the sum of squared deviations from the mean. It is continuous and strictly Schur concave under the equality-oriented convention of this paper.

Consider the zero-sum displacement

$$\mathbf{z} = \left(-\frac{1}{2}, 1, -\frac{1}{2}\right).$$

Apply this displacement first to

$$\mathbf{x} = \left(\frac{1}{2}, \frac{1}{2}, 5\right).$$

The resulting profile is

$$\mathbf{y} = \mathbf{x} + \mathbf{z} = \left(0, \frac{3}{2}, \frac{9}{2}\right).$$

Both profiles have mean 2, and direct calculation gives

$$I(\mathbf{x}) = -\frac{27}{2}, \qquad I(\mathbf{y}) = -\frac{21}{2}.$$

Hence $I(\mathbf{y}) > I(\mathbf{x})$, so the displacement is improving at the initial profile $\mathbf{x}$.

Now apply the same displacement to

$$\mathbf{x}' = \left(\frac{1}{2}, 2, \frac{7}{2}\right).$$

The resulting profile is

$$\mathbf{y}' = \mathbf{x}' + \mathbf{z} = (0, 3, 3).$$

These profiles also have mean 2, but now

$$I(\mathbf{x}') = -\frac{9}{2}, \qquad I(\mathbf{y}') = -6.$$

Hence $I(\mathbf{y}') < I(\mathbf{x}')$, so the same displacement is worsening at the initial profile $\mathbf{x}'$.

The two transitions generate exactly the same effect:

$$\mathbf{y} - \mathbf{x} = \mathbf{z} = \mathbf{y}' - \mathbf{x}'.$$

Nevertheless, the sign of the static change is positive at one baseline and negative at the other.

This reversal does not contradict Schur concavity. The second endpoint pair is not ordered by the Lorenz preorder, so Schur concavity imposes no ranking between its endpoints. More generally, Schur concavity governs comparisons covered by the Lorenz preorder; it does not assign a baseline-independent sign to every feasible displacement.

The example shows that continuity, nonconstancy, and strict Schur concavity do not by themselves guarantee the existence of a dynamic extension satisfying Static-Change Consistency and Transition-Profile Independence. The theorem below identifies the additional structure forced by such an extension: the static ranking must be representable, up to a strictly increasing transformation, by an ordered linear functional.

3.2 Fixed transformed aggregate

We begin with profiles whose transformed-coordinate levels have a common sum. Fix $b \in \mathbb{R}$ and define

$$V_{n,b} = \left\{ \mathbf{v} \in V_n : \sum_{i=1}^{n} v_i = b \right\}.$$

Every feasible transition $(\mathbf{v}, \mathbf{w})$ in this domain has a zero-sum transformed-coordinate effect, since

$$\sum_{i=1}^{n} (w_i - v_i) = 0.$$

For a transition $(\mathbf{v}, \mathbf{w}) \in V_{n,b} \times V_{n,b}$, define the cumulative transformed effect received by the poorest k ranks as

$$D_k(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^{k} (w_i - v_i), \qquad k \in \{1, \ldots, n-1\}.$$

These cumulative effects provide the fixed-transformed-aggregate form of Dynamic Lorenz comparisons. One transition is weakly more equalizing than another when its cumulative transformed effect is weakly larger at every nontrivial lower rank. The companion paper Carbonell-Nicolau (2026) formally defines and axiomatizes the corresponding Dynamic Lorenz dominance relation. Here it serves as a monotonicity benchmark for the dynamic extensions characterized below.

The set of transformed-coordinate effects that can arise from feasible transitions in this domain is

$$E_{n,b} = \left\{ \mathbf{w} - \mathbf{v} : \mathbf{v}, \mathbf{w} \in V_{n,b} \right\}.$$

We call $E_{n,b}$ the attainable effect set. Since every profile in $V_{n,b}$ has the same sum, every $\mathbf{z} \in E_{n,b}$ satisfies $\sum_{i=1}^{n} z_i = 0$. Hence $E_{n,b} \subseteq H_0^n$.

Theorem 2 (Fixed transformed-aggregate characterization)**.** *Let $I^* : V_{n,b} \to \mathbb{R}$ be continuous, nonconstant, and Schur concave. The following statements are equivalent.*

1. *There exists a dynamic index $J^* : V_{n,b} \times V_{n,b} \to \mathbb{R}$ satisfying Normalization, Static-Change Consistency, and Transition-Profile Independence.*

2. *There exist weights $q_1 \geq \cdots \geq q_n$, not all equal, and a strictly increasing function ϕ, defined on the range of*

$$G_{\mathbf{q}}(\mathbf{v}) = \sum_{i=1}^{n} q_i v_i,$$

such that $I^*(\mathbf{v}) = \phi\big(G_{\mathbf{q}}(\mathbf{v})\big)$ for every $\mathbf{v} \in V_{n,b}$.

When these equivalent conditions hold, every dynamic index satisfying the three axioms has the form $J^*(\mathbf{v}, \mathbf{w}) = F(\mathbf{w} - \mathbf{v})$, where $F : E_{n,b} \to \mathbb{R}$ satisfies $F(\mathbf{0}) = 0$ and

$$F(\mathbf{z}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0, \qquad \text{for every } \mathbf{z} \in E_{n,b}.$$

Conversely, every function F with these properties defines a dynamic index satisfying Normalization, Static-Change Consistency, and Transition-Profile Independence.

A linear member of this class is obtained by choosing $F = G_{\mathbf{q}}$. It is

$$J_0^*(\mathbf{v}, \mathbf{w}) = G_{\mathbf{q}}(\mathbf{w}) - G_{\mathbf{q}}(\mathbf{v}).$$

This extension is weakly monotone with respect to Dynamic Lorenz dominance in transformed coordinates. Specifically, if two feasible transitions $(\mathbf{v}, \mathbf{w})$ and $(\mathbf{v}', \mathbf{w}')$ satisfy

$$D_k(\mathbf{v}', \mathbf{w}') \geq D_k(\mathbf{v}, \mathbf{w}), \qquad \textit{for every } k \in \{1, \dots, n-1\},$$

then $J_0^*(\mathbf{v}', \mathbf{w}') \geq J_0^*(\mathbf{v}, \mathbf{w})$. If, in addition, $q_1 > \cdots > q_n$, every strict Dynamic Lorenz improvement receives a strictly larger value.

Proof. We first prove that the static representation in item 2 yields the dynamic indices described in the theorem. We then derive both the static and dynamic representations from the existence claim in item 1.

Item 2 implies item 1 and characterizes the dynamic class. Suppose that $I^* = \phi \circ G_{\mathbf{q}}$, where $q_1 \geq \cdots \geq q_n$ and ϕ is strictly increasing. Let $F : E_{n,b} \to \mathbb{R}$ satisfy $F(\mathbf{0}) = 0$ and

$$F(\mathbf{z}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0, \qquad \text{for every } \mathbf{z} \in E_{n,b},$$

and define $J^*(\mathbf{v}, \mathbf{w}) = F(\mathbf{w} - \mathbf{v})$.

Normalization follows from $F(\mathbf{0}) = 0$, and Transition-Profile Independence follows because J^* depends only on $\mathbf{w} - \mathbf{v}$. Moreover,

$$J^*(\mathbf{v}, \mathbf{w}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{w}) \geq G_{\mathbf{q}}(\mathbf{v}) \Leftrightarrow I^*(\mathbf{w}) \geq I^*(\mathbf{v}).$$

The final equivalence follows from the strict monotonicity of ϕ. Thus J^* satisfies Static-Change Consistency. This proves item 1 and shows that every function F described in the theorem produces an admissible dynamic index.

We next verify the additional claims about the linear member. Choosing $F = G_{\mathbf{q}}$ gives

$$J_0^*(\mathbf{v}, \mathbf{w}) = G_{\mathbf{q}}(\mathbf{w}) - G_{\mathbf{q}}(\mathbf{v}).$$

Because every effect in $E_{n,b}$ has zero sum, summation by parts gives

$$J_0^*(\mathbf{v}, \mathbf{w}) = \sum_{k=1}^{n-1} D_k(\mathbf{v}, \mathbf{w})(q_k - q_{k+1}).$$

The coefficients $q_k - q_{k+1}$ are nonnegative. Hence $D_k(\mathbf{v}', \mathbf{w}') \geq D_k(\mathbf{v}, \mathbf{w})$ for every $k < n$ implies

$$J_0^*(\mathbf{v}', \mathbf{w}') - J_0^*(\mathbf{v}, \mathbf{w}) = \sum_{k=1}^{n-1} \big[D_k(\mathbf{v}', \mathbf{w}') - D_k(\mathbf{v}, \mathbf{w})\big](q_k - q_{k+1}) \geq 0.$$

Thus the linear extension is weakly monotone under Dynamic Lorenz dominance. If the weights are strictly decreasing and the cumulative comparison is strict at some rank, at least one term in the preceding sum is strictly positive. The extension then assigns the strict Dynamic Lorenz improvement a strictly larger value.

Item 1 implies item 2 and restricts every admissible dynamic index. Suppose that J^* satisfies Normalization, Static-Change Consistency, and Transition-Profile Independence. Define a preorder on $V_{n,b}$ by $\mathbf{w} \succcurlyeq \mathbf{v}$ if and only if $I^*(\mathbf{w}) \geq I^*(\mathbf{v})$. This preorder is complete, transitive, continuous, and nonconstant by the maintained properties of I^*.

We verify the condition needed for the affine representation theorem. If $\mathbf{w} - \mathbf{v} = \mathbf{w}' - \mathbf{v}'$ for four profiles in $V_{n,b}$, then Transition-Profile Independence gives $J^*(\mathbf{v}, \mathbf{w}) = J^*(\mathbf{v}', \mathbf{w}')$. Static-Change Consistency therefore implies

$$I^*(\mathbf{w}) \geq I^*(\mathbf{v}) \Leftrightarrow I^*(\mathbf{w}') \geq I^*(\mathbf{v}').$$

Thus the preorder satisfies feasible-displacement independence.

The domain $V_{n,b}$ is a convex subset of the affine hyperplane $\sum_i v_i = b$ and has nonempty relative interior. Its displacement space is

$$H_0^n = \left\{ \mathbf{z} \in \mathbb{R}^n : \sum_{i=1}^n z_i = 0 \right\}.$$

By Theorem 1, there is a nonzero linear functional $L : H_0^n \to \mathbb{R}$ such that

$$I^*(\mathbf{w}) \geq I^*(\mathbf{v}) \Leftrightarrow L(\mathbf{w} - \mathbf{v}) \geq 0$$

for all $\mathbf{v}, \mathbf{w} \in V_{n,b}$.

Write $L(\mathbf{z}) = \sum_{i=1}^n q_i z_i$. To order the coefficients, choose a strictly ordered relative-interior profile and make a sufficiently small movement in the progressive-transfer direction $\mathbf{e}_j - \mathbf{e}_{j+1}$. Schur concavity makes this movement weakly improving, so $L(\mathbf{e}_j - \mathbf{e}_{j+1}) = q_j - q_{j+1} \geq 0$. Therefore $q_1 \geq \cdots \geq q_n$. The weights are not all equal, since equal weights would induce the zero functional on H_0^n.

The functionals L and $G_{\mathbf{q}}$ represent the same ordering on $V_{n,b}$. Consequently, I^* is constant on each level set of $G_{\mathbf{q}}$. Define $\phi(t) = I^*(\mathbf{v})$ whenever $G_{\mathbf{q}}(\mathbf{v}) = t$. This is well defined, and the common ordering represented by I^* and $G_{\mathbf{q}}$ makes ϕ strictly increasing. Hence $I^* = \phi \circ G_{\mathbf{q}}$, which proves item 2.

For completeness, the representing functional on H_0^n is unique up to a positive scalar. Two coefficient vectors represent the same functional on H_0^n precisely when they differ by a common additive constant. Thus the weights are unique up to positive rescaling and the addition of a common constant.

It remains to recover the form of the given dynamic index. By Transition-Profile Independence, $J^*(\mathbf{v}, \mathbf{w})$ depends only on the attainable effect $\mathbf{z} = \mathbf{w} - \mathbf{v}$. Define $F : E_{n,b} \to \mathbb{R}$ by $F(\mathbf{z}) = J^*(\mathbf{v}, \mathbf{w})$ whenever $\mathbf{z} = \mathbf{w} - \mathbf{v}$. This definition is independent of the transition chosen to realize $\mathbf{z}$. Normalization gives $F(\mathbf{0}) = 0$, while Static-Change Consistency and the static representation give

$$F(\mathbf{z}) \geq 0 \Leftrightarrow I^*(\mathbf{w}) \geq I^*(\mathbf{v}) \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0.$$

Hence every admissible dynamic index has the stated profile-functional form. ∎

Corollary 1 (Dynamic-Lorenz-consistent extensions)**.** *Suppose that the equivalent conditions of Theorem 2 hold, and let $J^*(\mathbf{v}, \mathbf{w}) = F(\mathbf{w} - \mathbf{v})$ be an admissible dynamic extension. Then J^* is*

weakly monotone under Dynamic Lorenz dominance if and only if F preserves the cumulative-effect order: for every $\mathbf{z}, \mathbf{z}' \in E_{n,b}$,

$$\sum_{i=1}^{k} z'_i \geq \sum_{i=1}^{k} z_i \quad \text{for every } k \in \{1, \ldots, n-1\} \Rightarrow F(\mathbf{z}') \geq F(\mathbf{z}).$$

The index J^* is Dynamic-Lorenz consistent, preserving both weak and strict comparisons, if and only if F preserves this weak order and satisfies $F(\mathbf{z}') > F(\mathbf{z})$ whenever the displayed cumulative inequalities hold and at least one of them is strict.

Proof. By Theorem 2, every admissible dynamic index depends on a transition only through its effect profile. Dynamic Lorenz dominance is precisely the cumulative-effect order displayed above. The result follows. ■

The theorem and the corollary impose different restrictions on the profile functional F. The theorem characterizes the full class of dynamic indices satisfying N, SC, and TPI. These axioms require $F(\mathbf{0}) = 0$ and $F(\mathbf{z}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0$, but they do not determine how F ranks arbitrary pairs of nonzero effect profiles. In particular, they do not require F to depend only on $G_{\mathbf{q}}(\mathbf{z})$, to be additive or linear, or to vanish whenever $G_{\mathbf{q}}(\mathbf{z}) = 0$.

Dynamic Lorenz monotonicity adds the requirement that F preserve the cumulative-effect order. Taking $F = G_{\mathbf{q}}$ yields a linear extension with this property because the rank weights are weakly decreasing. If they are strictly decreasing, the extension also preserves every strict Dynamic Lorenz comparison. With weakly decreasing weights, a strict improvement may leave its value unchanged when all strict cumulative inequalities occur at cuts for which $q_k = q_{k+1}$.

3.3 Variable transformed aggregate

We now allow the mean transformed-coordinate level to vary. For a transition $(\mathbf{v}, \mathbf{w}) \in V_n \times V_n$, define its cumulative centered effect by

$$\widetilde{D}_k(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^{k} \widetilde{\tau}_i(\mathbf{v}, \mathbf{w}), \qquad k \in \{1, \ldots, n-1\}.$$

The set of attainable centered effects is

$$\widetilde{E}_n = \{\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}) : \mathbf{v}, \mathbf{w} \in V_n\} \subseteq H_0^n.$$

Theorem 3 (Variable transformed-aggregate characterization)**.** *Let $I^* : V_n \to \mathbb{R}$ be continuous, nonconstant, and Schur concave. Suppose that I^* is translation invariant in transformed coordinates: $I^*(\mathbf{v}+c\mathbf{1}) = I^*(\mathbf{v})$ whenever both profiles belong to the maintained domain. The following statements are equivalent.*

1. *There exists a dynamic index $J^* : V_n \times V_n \to \mathbb{R}$ satisfying Normalization, Static-Change Consistency, and Centered Transition-Profile Independence.*

2. *There exist weights $q_1 \geq \cdots \geq q_n$ satisfying $\sum_{i=1}^n q_i = 0$, not all zero, and a strictly increasing function ϕ, defined on the range of $G_{\mathbf{q}}$, such that*

$$G_{\mathbf{q}}(\mathbf{v}) = \sum_{i=1}^{n} q_i v_i$$

and $I^*(\mathbf{v}) = \phi(G_{\mathbf{q}}(\mathbf{v}))$ for every $\mathbf{v} \in V_n$.

When these conditions hold, every dynamic index satisfying the three axioms has the form

$$J^*(\mathbf{v}, \mathbf{w}) = F\big(\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w})\big),$$

where $F : \widetilde{E}_n \to \mathbb{R}$ satisfies $F(\mathbf{0}) = 0$ and

$$F(\mathbf{z}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0, \qquad \text{for every } \mathbf{z} \in \widetilde{E}_n.$$

Conversely, every such function F defines an admissible dynamic index.

The linear choice $F = G_{\mathbf{q}}$ gives

$$J_0^*(\mathbf{v}, \mathbf{w}) = G_{\mathbf{q}}(\mathbf{w}) - G_{\mathbf{q}}(\mathbf{v}).$$

If two feasible transitions satisfy $\widetilde{D}_k(\mathbf{v}', \mathbf{w}') \geq \widetilde{D}_k(\mathbf{v}, \mathbf{w})$ for every $k < n$, then $J_0^*(\mathbf{v}', \mathbf{w}') \geq J_0^*(\mathbf{v}, \mathbf{w})$. If the weights are strictly decreasing, every strict centered Dynamic Lorenz improvement receives a strictly larger value.

Proof. We first derive the dynamic class from the static representation and then prove the converse.

Item 2 implies item 1 and characterizes the dynamic class. Suppose that $I^* = \phi \circ G_{\mathbf{q}}$, where ϕ is strictly increasing and the weights are weakly decreasing and sum to zero. Let $F : \widetilde{E}_n \to \mathbb{R}$ satisfy $F(\mathbf{0}) = 0$ and $F(\mathbf{z}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0$. Define $J^*(\mathbf{v}, \mathbf{w}) = F(\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}))$.

Normalization and Centered Transition-Profile Independence follow directly from this definition. Since the weights sum to zero, $G_{\mathbf{q}}(\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w})) = G_{\mathbf{q}}(\mathbf{w}) - G_{\mathbf{q}}(\mathbf{v})$. Therefore,

$$J^*(\mathbf{v}, \mathbf{w}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{w}) \geq G_{\mathbf{q}}(\mathbf{v}) \Leftrightarrow I^*(\mathbf{w}) \geq I^*(\mathbf{v}).$$

Thus J^* satisfies Static-Change Consistency.

For the linear choice $F = G_{\mathbf{q}}$, summation by parts gives

$$J_0^*(\mathbf{v}, \mathbf{w}) = \sum_{k=1}^{n-1} \widetilde{D}_k(\mathbf{v}, \mathbf{w})(q_k - q_{k+1}).$$

The coefficients are nonnegative, so centered cumulative dominance implies a weakly larger value. If the weights are strictly decreasing and at least one cumulative inequality is strict, the value is strictly larger.

Item 1 implies item 2 and restricts every admissible dynamic index. Suppose that J^* satisfies the three axioms. Let $C_n(\mathbf{v}) = \mathbf{v} - \bar{v}\mathbf{1}$ and define the centered domain $V_{n,0} = V_n \cap H_0^n$. Translation invariance gives $I^*(\mathbf{v}) = I^*(C_n(\mathbf{v}))$, so the static ranking is completely determined by its restriction to $V_{n,0}$.

On this centered domain, raw and centered effects coincide. Hence Centered Transition-Profile Independence becomes Transition-Profile Independence. The restriction of J^* and I^* to $V_{n,0}$ therefore satisfies the hypotheses of Theorem 2. That theorem yields weakly decreasing weights, not all equal, and a strictly increasing function ϕ such that $I^*(\mathbf{v}) = \phi(G_{\mathbf{q}}(\mathbf{v}))$ for every $\mathbf{v} \in V_{n,0}$.

The weights are identified on H_0^n only up to a common additive constant. Choose the unique representative satisfying $\sum_i q_i = 0$. Since the weights sum to zero, $G_{\mathbf{q}}(\mathbf{v}) = G_{\mathbf{q}}(C_n(\mathbf{v}))$ for every $\mathbf{v} \in V_n$. Because $C_n(\mathbf{v}) \in V_{n,0}$ and $V_{n,0} \subseteq V_n$, the ranges of $G_{\mathbf{q}}$ on V_n and on $V_{n,0}$ coincide. Translation invariance extends the representation to all of V_n:

$$I^*(\mathbf{v}) = I^*(C_n(\mathbf{v})) = \phi(G_{\mathbf{q}}(C_n(\mathbf{v}))) = \phi(G_{\mathbf{q}}(\mathbf{v})).$$

Finally, Centered Transition-Profile Independence implies that $J^*(\mathbf{v}, \mathbf{w})$ depends only on $\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w})$. Thus there is a well-defined function $F : \widetilde{E}_n \to \mathbb{R}$ such that $J^*(\mathbf{v}, \mathbf{w}) = F(\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w}))$. Normalization gives $F(\mathbf{0}) = 0$, while Static-Change Consistency and the static representation give

$$F(\mathbf{z}) \geq 0 \Leftrightarrow G_{\mathbf{q}}(\mathbf{z}) \geq 0, \qquad \text{for every } \mathbf{z} \in \widetilde{E}_n.$$

This proves the stated characterization. ■

The same distinction applies when the transformed aggregate varies. The theorem characterizes all baseline-independent extensions by specifying exactly which centered effect profiles receive nonnegative values. Among them, an index $J^*(\mathbf{v}, \mathbf{w}) = F\big(\widetilde{\boldsymbol{\tau}}(\mathbf{v}, \mathbf{w})\big)$ is weakly monotone under centered Dynamic Lorenz dominance if and only if F preserves the centered cumulative-effect order on $\widetilde{E}_n$. It is centered Dynamic Lorenz consistent if and only if F also assigns a strictly larger value whenever the centered cumulative inequalities are weakly satisfied at every rank and strictly satisfied at some rank.

Taking $F = G_{\mathbf{q}}$ yields a linear extension that is weakly monotone because the weights are weakly decreasing. If the weights are strictly decreasing, every strict centered Dynamic Lorenz comparison produces a strictly larger value, so the linear extension is centered Dynamic Lorenz consistent. With weakly decreasing weights, a strict improvement may leave its value unchanged when all strict cumulative inequalities occur at cuts for which $q_k = q_{k+1}$.

4 Absolute and relative applications

We now derive the absolute and relative results by using identity and logarithmic coordinates, respectively.

4.1 Absolute geometry: identity coordinates

Let $\psi_A(t) = t$. The transformed economic domain is

$$V_n^+ = \{\mathbf{v} \in V_n : v_1 \geq 0\}.$$

Corollary 2 (Fixed arithmetic mean)**.** *On the domain of ordered nonnegative distributions with a fixed positive arithmetic mean, a continuous, nonconstant, and Schur-concave static index admits a dynamic index satisfying N, SC, and TPI if and only if*

$$I^A(\mathbf{x}) = \phi_A\left(\sum_{i=1}^{n} q_i^A x_i\right)$$

for some weakly decreasing weights $q_1^A \geq \cdots \geq q_n^A$, not all equal, and some strictly increasing function ϕ_A. The weights are unique up to positive rescaling and the addition of a common constant.

When these conditions hold, every such dynamic index has the form

$$J^A(\mathbf{x}, \mathbf{y}) = F(\mathbf{y} - \mathbf{x}),$$

where F is defined on the attainable absolute effect profiles, $F(\mathbf{0}) = 0$, and $F(\mathbf{z}) \geq 0 \Leftrightarrow \sum_i q_i^A z_i \geq 0$ for every attainable effect $\mathbf{z}$. Conversely, every such function F defines an admissible dynamic index.

The linear choice is

$$J_0^A(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} q_i^A (y_i - x_i).$$

It is weakly monotone under absolute Dynamic Lorenz dominance and, when the weights are strictly decreasing, assigns a strictly larger value to every strict absolute Dynamic Lorenz improvement.

Proof. Let

$$D = \left\{\mathbf{x} \in V_n : x_1 \geq 0,\ \sum_{i=1}^{n} x_i = b\right\},$$

where $b > 0$ is the fixed aggregate. This domain is convex, has nonempty relative interior in the fixed-sum hyperplane, and satisfies

$$\operatorname{span}(D - D) = H_0^n.$$

The proof of Theorem 2 applies to D: Theorem 1 yields a nonzero linear functional on H_0^n, and Schur concavity orders its coefficients by the same progressive-transfer argument. The characterization of the admissible dynamic indices and the Dynamic Lorenz claims then follow exactly as in Theorem 2. The domain includes ties and zero-income boundary profiles. ∎

Corollary 3 (Variable arithmetic means)**.** *On the ordered nonnegative domain, let I^A be continuous, nonconstant, Schur concave, and invariant under every admissible common translation. It admits a dynamic index satisfying N, SC, and CTPI if and only if*

$$I^A(\mathbf{x}) = \phi_A\left(\sum_{i=1}^{n} q_i^A x_i\right), \qquad q_1^A \geq \cdots \geq q_n^A, \qquad \sum_{i=1}^{n} q_i^A = 0,$$

where the weights are not all zero and ϕ_A is strictly increasing.

When these conditions hold, every such dynamic index has the form

$$J^A(\mathbf{x}, \mathbf{y}) = F\big(C_n(\mathbf{y}) - C_n(\mathbf{x})\big),$$

where F is defined on the attainable centered absolute effect profiles, $F(\mathbf{0}) = 0$, and $F(\mathbf{z}) \geq 0 \Leftrightarrow \sum_i q_i^A z_i \geq 0$ for every attainable centered effect $\mathbf{z}$. Conversely, every such function F defines an admissible dynamic index.

The linear choice is

$$J_0^A(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} q_i^A (y_i - x_i).$$

It is weakly monotone under centered absolute Dynamic Lorenz dominance and, when the weights are strictly decreasing, assigns a strictly larger value to every strict centered absolute Dynamic Lorenz improvement.

Proof. Let

$$V_n^+ = \{\mathbf{v} \in V_n : v_1 \geq 0\}.$$

We extend the static and dynamic indices from V_n^+ to V_n. For each $\mathbf{v} \in V_n$, choose $c_{\mathbf{v}} \in \mathbb{R}$ such that $\mathbf{v} + c_{\mathbf{v}}\mathbf{1} \in V_n^+$, and define $\widetilde{I}^A(\mathbf{v}) = I^A(\mathbf{v} + c_{\mathbf{v}}\mathbf{1})$. This definition is independent of the chosen admissible shift. Indeed, if both $c_{\mathbf{v}}$ and $c'_{\mathbf{v}}$ are admissible, translation invariance gives $I^A(\mathbf{v} + c_{\mathbf{v}}\mathbf{1}) = I^A(\mathbf{v} + c'_{\mathbf{v}}\mathbf{1})$. For $\mathbf{v}, \mathbf{w} \in V_n$, choose admissible shifts $c_{\mathbf{v}}$ and $c_{\mathbf{w}}$ and define

$$\widetilde{J}^A(\mathbf{v}, \mathbf{w}) = J^A\big(\mathbf{v} + c_{\mathbf{v}}\mathbf{1}, \mathbf{w} + c_{\mathbf{w}}\mathbf{1}\big).$$

This definition is also independent of the chosen shifts. Centering is unaffected by a common translation of either endpoint, so any two admissibly shifted versions of the transition have the same centered effect profile. CTPI then assigns them the same value.

The index $\widetilde{I}^A$ is continuous. To verify continuity at $\mathbf{v} \in V_n$, choose one constant c sufficiently large that $\mathbf{v}' + c\mathbf{1} \in V_n^+$ for every $\mathbf{v}'$ in a neighborhood of $\mathbf{v}$. On that neighborhood, $\widetilde{I}^A(\mathbf{v}') = I^A(\mathbf{v}' + c\mathbf{1})$, so continuity follows from continuity of I^A. Nonconstancy is inherited from the restriction to V_n^+.

The extension $\widetilde{I}^A$ is Schur concave. If $\mathbf{v}, \mathbf{w} \in V_n$ have the same sum and $\mathbf{v}$ Lorenz-dominates $\mathbf{w}$, choose a common upward shift c that makes both profiles nonnegative. Common translation preserves their Lorenz comparison, and Schur concavity of I^A gives $\widetilde{I}^A(\mathbf{v}) \geq \widetilde{I}^A(\mathbf{w})$. Translation invariance of $\widetilde{I}^A$ follows immediately from its definition.

The extended dynamic index satisfies N. Any shifted version of the null transition $(\mathbf{v}, \mathbf{v})$ has the same centered effect profile as a null transition in V_n^+, so CTPI and N for J^A give $\widetilde{J}^A(\mathbf{v}, \mathbf{v}) = 0$. It satisfies SC because, for any admissible endpoint shifts,

$$\widetilde{J}^A(\mathbf{v}, \mathbf{w}) \geq 0 \Leftrightarrow I^A(\mathbf{w} + c_{\mathbf{w}}\mathbf{1}) \geq I^A(\mathbf{v} + c_{\mathbf{v}}\mathbf{1}) \Leftrightarrow \widetilde{I}^A(\mathbf{w}) \geq \widetilde{I}^A(\mathbf{v}).$$

It also satisfies CTPI. If two transitions in V_n have the same centered effect profile, any admissibly shifted versions of those transitions in V_n^+ retain that same centered effect profile. CTPI for J^A then gives equality of their extended values.

The pair $(\widetilde{I}^A, \widetilde{J}^A)$ satisfies all the hypotheses of Theorem 3. That theorem yields weakly decreasing weights $q_1^A \geq \cdots \geq q_n^A$, $\sum_{i=1}^n q_i^A = 0$, not all zero, and a strictly increasing function ϕ_A such that

$$\widetilde{I}^A(\mathbf{v}) = \phi_A\left(\sum_{i=1}^{n} q_i^A v_i\right), \qquad \text{for every } \mathbf{v} \in V_n.$$

It also gives the stated characterization of all extended dynamic indices, including the linear member and its centered Dynamic Lorenz properties. Restricting these conclusions to V_n^+ proves the necessity claims of the corollary.

Conversely, suppose that the stated representation holds on V_n^+ and let F satisfy the conditions in the corollary. Define $J^A(\mathbf{x}, \mathbf{y}) = F\big(C_n(\mathbf{y}) - C_n(\mathbf{x})\big)$. The same argument as in Theorem 3 shows that J^A satisfies N, SC, and CTPI. The linear choice and the centered Dynamic Lorenz claims also follow from that theorem. ■

4.2 Relative geometry: logarithmic coordinates

Let $\psi_R(t) = \log t$ on strictly positive incomes. The logarithmic map sends the ordered positive domain onto V_n. Endpoint incomes must be positive, but the log effect $\log y_i - \log x_i$ may be positive, zero, or negative. The relative framework therefore permits income losses but excludes zero or negative endpoint incomes.

Corollary 4 (Fixed geometric mean)**.** *On the domain of ordered positive distributions with a fixed geometric mean, a continuous and nonconstant static index whose log-coordinate representation is Schur concave admits a dynamic index satisfying N, SC, and TPI in logarithmic coordinates if and only if*

$$I^R(\mathbf{x}) = \phi_R\left(\sum_{i=1}^n q_i^R \log x_i\right)$$

for some weakly decreasing weights $q_1^R \geq \cdots \geq q_n^R$, not all equal, and some strictly increasing function ϕ_R. The weights are unique up to positive rescaling and the addition of a common constant.

When these conditions hold, every such dynamic index has the form

$$J^R(\mathbf{x}, \mathbf{y}) = F\big(\log \mathbf{y} - \log \mathbf{x}\big),$$

where F is defined on the attainable log-effect profiles, $F(\mathbf{0}) = 0$, and $F(\mathbf{z}) \geq 0 \Leftrightarrow \sum_i q_i^R z_i \geq 0$ for every attainable effect $\mathbf{z}$. Conversely, every such function F defines an admissible dynamic index.

The linear choice is

$$J_0^R(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n q_i^R(\log y_i - \log x_i).$$

It is weakly monotone under relative Dynamic Lorenz dominance and, when the weights are strictly decreasing, assigns a strictly larger value to every strict relative Dynamic Lorenz improvement.

Proof. For geometric mean g, the logarithmic map is an order-preserving bijection onto $V_{n,n\log g}$. Apply Theorem 2. ■

Corollary 5 (Variable geometric means)**.** *On the ordered positive domain, let I^R be continuous, nonconstant, and scale invariant, and suppose that its log-coordinate representation is Schur concave. It admits a dynamic index satisfying N, SC, and CTPI in logarithmic coordinates if and only if*

$$I^R(\mathbf{x}) = \phi_R\left(\sum_{i=1}^n q_i^R \log x_i\right), \qquad q_1^R \geq \cdots \geq q_n^R, \qquad \sum_{i=1}^n q_i^R = 0,$$

where the weights are not all zero and ϕ_R is strictly increasing.

When these conditions hold, every such dynamic index has the form

$$J^R(\mathbf{x}, \mathbf{y}) = F\big(C_n(\log \mathbf{y}) - C_n(\log \mathbf{x})\big),$$

where F is defined on the attainable centered log-effect profiles, $F(\mathbf{0}) = 0$, and $F(\mathbf{z}) \geq 0 \Leftrightarrow \sum_i q_i^R z_i \geq 0$ for every attainable centered effect $\mathbf{z}$. Conversely, every such function F defines an admissible dynamic index.

The linear choice is

$$J_0^R(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} q_i^R (\log y_i - \log x_i).$$

It is weakly monotone under centered relative Dynamic Lorenz dominance and, when the weights are strictly decreasing, assigns a strictly larger value to every strict centered relative Dynamic Lorenz improvement.

Proof. Under $\mathbf{v} = \log \mathbf{x}$, scale invariance becomes translation invariance because $\log(\lambda \mathbf{x}) = \log \mathbf{x} + (\log \lambda)\mathbf{1}$. Apply Theorem 3. Tied positive incomes are admissible, but zero-income profiles remain excluded because their logarithmic coordinates are not finite. ■

5 Relation to cardinal welfare-change measurement

Bossert and Dutta (2019) study measures of welfare change of the form

$$V(\mathbf{x}^0, \mathbf{x}^1) = W(\mathbf{x}^1) - W(\mathbf{x}^0).$$

This welfare-difference requirement is central to their framework: W has cardinal significance, so values of V can be compared across transitions. They show that cumulative dominance of ranked income differences characterizes unanimous comparisons over generalized-Gini welfare differences. They also characterize those differences using welfare-difference compatibility, strict monotonicity, strict S-concavity, positive linear homogeneity, and weak independence of income source. For distributions with a common mean, they obtain a corresponding result for changes in generalized-Gini inequality indices.

The present paper asks a different representation question. It does not assume that a dynamic index is the difference of cardinal state values. Instead, it asks which ordinal static rankings admit some dynamic index satisfying Static-Change Consistency and profile independence. The dynamic index need not measure the magnitude of static change. It must be nonnegative exactly when the static change is weakly improving. Its value must also depend on the relevant effect profile rather than separately on the baseline. The main result is that the existence of such an extension already forces the static ranking to be a strictly increasing transformation of a generalized-Gini functional. Thus generalized-Gini geometry is derived from dynamic extendability without assuming welfare-difference compatibility, positive linear homogeneity, or weak independence of income source.

This distinction also separates the conclusions about dynamic measurement. In the Bossert and Dutta (2019) framework, the dynamic measure is the generalized-Gini difference itself. Without welfare-difference compatibility and the accompanying cardinal restrictions, the admissible dynamic indices characterized here have a more general functional form: they may be general functions of the relevant effect profile, provided they are nonnegative exactly when the corresponding generalized-Gini difference is nonnegative. The generalized-Gini difference is a linear member of this class and provides the point of contact between the two frameworks. The present paper does not, however, claim that the two characterizations are nested, since their primitives and characterized objects differ.

The domains and interpretations also differ. Bossert and Dutta work with ranked income differences for a fixed population and allow aggregate income change to enter welfare change. The present paper isolates pure inequality change by centering transformed-coordinate effects when the aggregate varies. Identity coordinates remove uniform additive changes, while logarithmic coordinates remove proportional scaling and yield a relative, scale-invariant characterization. The analysis also covers variable populations: independent replication of the two endpoints makes the finite-population rank weights agree proportionally under replication. After a consistent normalization, these weights generate a common weighting function on population shares. Bossert and Dutta identify replication as a possible route to comparisons across population sizes, whereas the present paper develops that route as a formal characterization.

The relationship can be summarized as follows. Bossert and Dutta (2019) characterize cardinal generalized-Gini differences and the dominance comparisons they support. The present paper characterizes the ordinal static rankings that admit a baseline-independent dynamic index. Such an index evaluates a transition as weakly improving exactly when the static ranking places the final distribution weakly above the initial distribution. The linear extension belongs to the generalized-Gini difference class, but the existence result, the centered absolute and relative geometries, and the variable-population characterization are distinct contributions.

6 Variable population

We now allow the endpoints of a transition to have different population sizes. The fixed-population results provide a decreasing zero-sum rank-weight vector at each population size, but they do not relate the vectors obtained at different sizes. This section shows how replication consistency supplies that relation.

The argument has three main steps. First, distributions with different population sizes are replicated to a common size, allowing their centered effect profiles to be compared rank by rank. Second, replication consistency requires the weights assigned to the replicated copies of an original rank to add up proportionally to the original weight. Because each fixed-population weight vector is determined only up to positive rescaling, this agreement is initially proportional. The vectors can then be normalized so that the weights assigned to the finer ranks replacing an original rank sum exactly to its original weight. Third, the resulting weight vectors are shown to be the increments of a single concave function on population shares. Centered-quantile continuity extends this function continuously to $[0, 1]$.

The argument applies to both geometries considered above. In the absolute case, the weights act on income levels; in the relative case, they act on log incomes. In either case, the difference between the weighted endpoint values is independent of the common population size used to represent the endpoints and preserves the corresponding unrestricted Dynamic Lorenz order.

This construction differs from imposing a static population principle directly. It uses the index-valued counterpart of the Dynamic Population Principle in Carbonell-Nicolau (2026): independently replicating the two endpoints does not change the evaluation of a transition. In that paper, the principle extends Dynamic Lorenz dominance to variable populations. Here, it links the rank-weight vectors obtained at different dimensions and yields a common weighting function on population shares. The construction is also related to the replication approaches of Shorrocks (1983) and Ebert (1988), but the cross-population restriction is derived from dynamic replication rather than imposed

directly on the static ranking.

For each $n \geq 2$, let

$$\mathbf{S}_n = \{\mathbf{x} \in \mathbb{R}^n : 0 \leq x_1 \leq \cdots \leq x_n\}, \qquad \mathbf{S}_n^{++} = \{\mathbf{x} \in \mathbb{R}^n : 0 < x_1 \leq \cdots \leq x_n\}.$$

Thus zero incomes are admitted in the absolute domain and excluded in the relative domain. Write $\mathbf{S} = \bigcup_{n\geq 2} \mathbf{S}_n$ and $\mathbf{S}^{++} = \bigcup_{n\geq 2} \mathbf{S}_n^{++}$, and define $\mathcal{T}^A = \mathbf{S} \times \mathbf{S}$ and $\mathcal{T}^R = \mathbf{S}^{++} \times \mathbf{S}^{++}$.

The static indices $I^A : \mathbf{S} \to \mathbb{R}$ and $I^R : \mathbf{S}^{++} \to \mathbb{R}$ are single functions on their respective union domains, not separate indices defined and scaled independently at each population size. Similarly, the dynamic indices $J^A : \mathcal{T}^A \to \mathbb{R}$ and $J^R : \mathcal{T}^R \to \mathbb{R}$ evaluate all admissible transitions, including those whose endpoints have different population sizes. Static-Change Consistency requires the dynamic evaluation of a cross-population transition to agree with the static comparison of its endpoints. The analysis below derives a common ordinal representation across population sizes rather than combining unrelated fixed-population representations.

If $\mathbf{x} \in \mathbf{S}_n$ and $m \geq 1$, its m-replica is

$$\mathbf{x}^{[m]} = (\underbrace{x_1, \ldots, x_1}_{m \text{ times}}, \ldots, \underbrace{x_n, \ldots, x_n}_{m \text{ times}}) \in \mathbf{S}_{mn}.$$

The same notation is used on the positive-income domain.

6.1 Common-refinement transition profiles

Consider a transition $(\mathbf{x}, \mathbf{y}) \in \mathcal{T}^K$, where $K \in \{A, R\}$, and suppose its endpoints have population sizes n_x and n_y. Choose a common multiple N of these sizes and form $\widetilde{\mathbf{x}} = \mathbf{x}^{[N/n_x]}$ and $\widetilde{\mathbf{y}} = \mathbf{y}^{[N/n_y]}$.

The centered absolute and relative effect profiles at population N are

$$\widetilde{\boldsymbol{\tau}}_N^A(\mathbf{x}, \mathbf{y}) = C_N(\widetilde{\mathbf{y}}) - C_N(\widetilde{\mathbf{x}})$$

and

$$\widetilde{\boldsymbol{\tau}}_N^R(\mathbf{x}, \mathbf{y}) = C_N(\log \widetilde{\mathbf{y}}) - C_N(\log \widetilde{\mathbf{x}}),$$

respectively. Equivalently, the relative profile can be written as $\log(\widetilde{\mathbf{y}}/\mu_G(\widetilde{\mathbf{y}})) - \log(\widetilde{\mathbf{x}}/\mu_G(\widetilde{\mathbf{x}}))$, with logarithms and quotients taken coordinatewise. Thus relative endpoint normalization is simply centering in log-income coordinates.

For $k \in \{0, \ldots, N\}$, define the normalized cumulative effect by

$$C_N^K\left(\frac{k}{N}; \mathbf{x}, \mathbf{y}\right) = \frac{1}{N} \sum_{i=1}^{k} \widetilde{\tau}_{N,i}^K(\mathbf{x}, \mathbf{y}),$$

using an empty sum at $k = 0$, and extend the function linearly between grid points. Centering gives $C_N^K(0; \mathbf{x}, \mathbf{y}) = C_N^K(1; \mathbf{x}, \mathbf{y}) = 0$.

Lemma 3 (Invariance under further replication)**.** *For every positive integer m and every $p \in [0, 1]$,*

$$C_{mN}^K\left(p; \mathbf{x}^{[m]}, \mathbf{y}^{[m]}\right) = C_N^K(p; \mathbf{x}, \mathbf{y}).$$

Proof. At the grid point $k/N = mk/(mN)$, each of the first k effects is repeated m times. Hence the numerator and denominator in the normalized cumulative sum are both multiplied by m, leaving its value unchanged. The additional grid points lie on the same linear segments because each original effect is repeated. Therefore, the two curves coincide on all of $[0, 1]$. ∎

The lemma shows that replication to a common population does not alter the cumulative-effect curve. Consequently, we may write $C^K(\,\cdot\,; \mathbf{x}, \mathbf{y})$ without specifying the common refinement.

Definition 5 (Unrestricted Dynamic Lorenz dominance)**.** For $K \in \{A, R\}$, write

$$(\mathbf{x}', \mathbf{y}') \succcurlyeq^K_{\mathrm{UDL}} (\mathbf{x}, \mathbf{y}) \Leftrightarrow C^K(p; \mathbf{x}', \mathbf{y}') \geq C^K(p; \mathbf{x}, \mathbf{y}), \quad \text{for every } p \in [0, 1].$$

This is the unrestricted Dynamic Lorenz order developed in Carbonell-Nicolau (2026), expressed through the normalized cumulative-effect curves used here. At any common refinement N, it is enough to compare the curves at the grid points k/N, $k \in \{1, \ldots, N-1\}$.

Definition 6 (Dynamic Population Principle)**.** A dynamic index J satisfies the *Dynamic Population Principle* (DPP) if

$$J(\mathbf{x}^{[r]}, \mathbf{y}^{[s]}) = J(\mathbf{x}, \mathbf{y})$$

for every admissible transition $(\mathbf{x}, \mathbf{y})$ and all positive integers r and s.

DPP permits the two endpoints to be replicated independently. It treats replication as a change in population representation, not as a change in the distributive content of the transition. This is stronger than requiring invariance only when both endpoints are replicated by the same factor.

Lemma 4 (Dynamic replication implies static replication invariance)**.** *Let I^* be a static index on the relevant union domain, and suppose that J^* satisfies N, SC, and DPP. Then, for every positive integer m,*

$$I^*(\mathbf{y}) \geq I^*(\mathbf{x}) \Leftrightarrow I^*(\mathbf{y}^{[m]}) \geq I^*(\mathbf{x}^{[m]}),$$

and $I^*(\mathbf{x}^{[m]}) = I^*(\mathbf{x})$ for every $\mathbf{x}$.

Proof. SC and common replication give

$$I^*(\mathbf{y}) \geq I^*(\mathbf{x}) \Leftrightarrow J^*(\mathbf{x}, \mathbf{y}) \geq 0 \Leftrightarrow J^*(\mathbf{x}^{[m]}, \mathbf{y}^{[m]}) \geq 0 \Leftrightarrow I^*(\mathbf{y}^{[m]}) \geq I^*(\mathbf{x}^{[m]}).$$

This proves preservation of static comparisons.

For numerical invariance, DPP and N give $J^*(\mathbf{x}, \mathbf{x}^{[m]}) = J^*(\mathbf{x}^{[m]}, \mathbf{x}^{[m]}) = 0$. Therefore, SC yields $I^*(\mathbf{x}^{[m]}) \geq I^*(\mathbf{x})$. Applying the same argument to the reverse transition gives the opposite inequality. Hence $I^*(\mathbf{x}^{[m]}) = I^*(\mathbf{x})$. ∎

Definition 7 (Unrestricted Centered Transition-Profile Independence)**.** For $K \in \{A, R\}$, let $J^* : \mathcal{T}^K \to \mathbb{R}$ be a dynamic index. It satisfies *K-Unrestricted Centered Transition-Profile Independence* (K-UCTPI) if, for any $(\mathbf{x}, \mathbf{y}), (\mathbf{x}', \mathbf{y}') \in \mathcal{T}^K$,

$$\widetilde{\boldsymbol{\tau}}^K_N(\mathbf{x}, \mathbf{y}) = \widetilde{\boldsymbol{\tau}}^K_N(\mathbf{x}', \mathbf{y}') \Rightarrow J^*(\mathbf{x}, \mathbf{y}) = J^*(\mathbf{x}', \mathbf{y}'),$$

where N is any common multiple of the four endpoint population sizes and the centered effect profiles are evaluated at population N.

By Lemma 3, the condition is unchanged under any further common refinement. DPP and UCTPI play different roles: DPP identifies a transition with its endpoint replicas, whereas UCTPI identifies transitions with the same centered effect profile at a common population size.

6.2 Agreement of rank weights across populations

At each population size n, the fixed-population results provide a decreasing zero-sum rank-weight vector $q_K^{(n)}$, determined only up to positive rescaling. We now ask how the vectors obtained at different population sizes must be related.

An unrestricted transition may have endpoints of different sizes. After replicating both endpoints to a common population size n, consider a further common replication by a factor m. Each rank at population n then becomes a block of m consecutive ranks at population mn. DPP requires the transition to retain its value under this further replication. Consequently, the total weight assigned to each replicated block must agree, up to a common positive factor, with the weight assigned to the corresponding rank at population n. Only the common-replication case of DPP is needed for this argument.

For $n \geq 2$ and $m \geq 1$, define $B_{n,m} : \mathbb{R}^{mn} \to \mathbb{R}^n$ by

$$\big(B_{n,m}\mathbf{q}\big)_i = \sum_{r=1}^{m} q_{(i-1)m+r}, \qquad i \in \{1, \dots, n\}.$$

Thus $B_{n,m}\mathbf{q}$ records the total weight assigned to the m replicated copies of each original rank.

Lemma 5 (Proportional agreement under replication)**.** *Fix $K \in \{A, R\}$. For every $n \geq 2$, let $q_K^{(n)} \in H_0^n$ be a nonzero decreasing vector representing the static ranking at population n. If the dynamic index satisfies SC and DPP, then, for every $n \geq 2$ and $m \geq 1$, there is a unique $\lambda_{n,m}^K > 0$ such that*

$$B_{n,m} q_K^{(mn)} = \lambda_{n,m}^K q_K^{(n)}.$$

Moreover,

$$\lambda_{n,m\ell}^K = \lambda_{n,m}^K \lambda_{mn,\ell}^K \qquad \text{for every } m, \ell \geq 1.$$

Proof. We first prove proportional agreement. Fix $\mathbf{v} \in H_0^n$. Choose a strictly ordered interior baseline at population n and a sufficiently small feasible transition whose centered effect is $\varepsilon\mathbf{v}$. In the relative case, make this construction in log-income coordinates. By the fixed-population representation and SC, the transition is weakly improving exactly when $q_K^{(n)} \cdot \mathbf{v} \geq 0$.

Replicate both endpoints by m. Their common population size becomes mn, their centered effect becomes $\varepsilon\mathbf{v}^{[m]}$, and DPP preserves the dynamic value. Applying SC and the fixed-population representation at population mn gives

$$q_K^{(n)} \cdot \mathbf{v} \geq 0 \Leftrightarrow q_K^{(mn)} \cdot \mathbf{v}^{[m]} \geq 0 \Leftrightarrow \big(B_{n,m} q_K^{(mn)}\big) \cdot \mathbf{v} \geq 0.$$

Thus $q_K^{(n)}$ and $B_{n,m} q_K^{(mn)}$ define the same oriented half-space on H_0^n. Both are nonzero on this space, so there is a unique $\lambda_{n,m}^K > 0$ such that $B_{n,m} q_K^{(mn)} = \lambda_{n,m}^K q_K^{(n)}$.

We next prove the relation among the proportionality factors. Replicating each rank m times and then each resulting rank ℓ times creates the same $m\ell$ copies as replicating the original rank $m\ell$ times at once. Hence $B_{n,m\ell} = B_{n,m} \circ B_{mn,\ell}$.

Apply this identity to $q_K^{(mn\ell)}$. Using proportional agreement first from population $mn\ell$ to mn and then from mn to n gives the first expression below; applying it directly from population $mn\ell$ to n gives the second:

$$B_{n,m\ell} q_K^{(mn\ell)} = \lambda_{mn,\ell}^K \lambda_{n,m}^K q_K^{(n)} = \lambda_{n,m\ell}^K q_K^{(n)}.$$

Since $q_K^{(n)} \neq \mathbf{0}$, the coefficients are equal. Therefore $\lambda^K_{n,m\ell} = \lambda^K_{n,m}\lambda^K_{mn,\ell}$. ∎

The agreement is only proportional because each fixed-population weight vector is determined up to positive rescaling. The next result shows that these scales can be chosen consistently so that, after normalization, the weights assigned to the finer ranks replacing an original rank sum exactly to its original weight.

Lemma 6 (Exact normalization across populations)**.** *Suppose that nonzero vectors $\{q^{(n)}\}_{n\geq 2}$ satisfy $B_{n,m}q^{(mn)} = \lambda_{n,m}q^{(n)}$ for positive multipliers such that*

$$\lambda_{n,m\ell} = \lambda_{n,m}\lambda_{mn,\ell}.$$

Then there are positive constants $\{a_n\}_{n\geq 2}$ such that $\widehat{q}^{(n)} = a_n q^{(n)}$ satisfies

$$B_{n,m}\widehat{q}^{(mn)} = \widehat{q}^{(n)}, \qquad \text{for every } n \geq 2 \text{ and } m \geq 1.$$

This normalized family is unique up to one common positive scalar.

Proof. We first construct the normalization. Fix a reference population $n_0 \geq 2$ and choose $a_{n_0} > 0$. For each $n \geq 2$, choose a common multiple N of n and n_0 and define

$$a_n = a_{n_0}\frac{\lambda_{n,N/n}}{\lambda_{n_0,N/n_0}}.$$

We first show that a_n does not depend on the chosen common multiple. Let N' be another common multiple of n and n_0, and choose a common multiple M of N and N'. Applying the multiplicative relation first through N gives $\lambda_{n,M/n} = \lambda_{n,N/n}\lambda_{N,M/N}$ and $\lambda_{n_0,M/n_0} = \lambda_{n_0,N/n_0}\lambda_{N,M/N}$. Hence

$$\frac{\lambda_{n,M/n}}{\lambda_{n_0,M/n_0}} = \frac{\lambda_{n,N/n}}{\lambda_{n_0,N/n_0}}.$$

Repeating the same argument through N' shows that this ratio also equals $\lambda_{n,N'/n}/\lambda_{n_0,N'/n_0}$. Thus either choice produces the same a_n, so the normalization is well defined.

We next prove exact agreement. Fix $n \geq 2$ and $m \geq 1$, and choose a common multiple N of mn and n_0. The definition of the normalizing constants gives

$$a_{mn}\lambda_{n,m} = a_{n_0}\frac{\lambda_{mn,N/(mn)}\lambda_{n,m}}{\lambda_{n_0,N/n_0}} = a_{n_0}\frac{\lambda_{n,N/n}}{\lambda_{n_0,N/n_0}} = a_n,$$

where the middle equality follows from the multiplicative relation for the λ's. Therefore,

$$B_{n,m}\widehat{q}^{(mn)} = a_{mn}B_{n,m}q^{(mn)} = a_{mn}\lambda_{n,m}q^{(n)} = \widehat{q}^{(n)}.$$

Thus the normalized weights satisfy the stated replication identity.

It remains to prove uniqueness. Suppose that $\widetilde{q}^{(n)} = \widetilde{a}_n q^{(n)}$ is another normalized family satisfying the same replication identity. Applying that agreement to populations n and mn gives $\widetilde{a}_{mn}\lambda_{n,m} = \widetilde{a}_n$. The normalization constructed above similarly gives $a_{mn}\lambda_{n,m} = a_n$. Dividing these identities yields

$$\frac{\widetilde{a}_{mn}}{a_{mn}} = \frac{\widetilde{a}_n}{a_n}.$$

Thus the ratio between the two normalizations is unchanged under replication. For any two population sizes n and r, choose a common multiple N. Applying the preceding identity along the replications from n to N and from r to N gives $\widetilde{a}_n/a_n = \widetilde{a}_N/a_N = \widetilde{a}_r/a_r$. Therefore the ratio is constant across population sizes, and the two normalized families differ by one common positive scalar. ■

After normalization, subdividing a rank does not change its total weight. The weights assigned to the resulting finer ranks may differ, but together they equal the original rank weight. The next subsection uses this property to construct a common weighting function across population sizes.

6.3 A population-share weighting function

The preceding normalization ensures that subdividing a rank preserves its total weight. We now show that the consistently normalized finite-population weights are the increments of a single function on population shares.

Lemma 7 (Cumulative weights on rational population shares)**.** *Let $\{\widehat{q}^{(n)}\}_{n\geq 2}$ be decreasing zero-sum vectors satisfying*

$$\sum_{r=1}^{m} \widehat{q}^{(mn)}_{(i-1)m+r} = \widehat{q}^{(n)}_i$$

for every $n \geq 2$, $m \geq 1$, and $i \in \{1, \dots, n\}$. Then there is a concave function $Q : \mathbb{Q} \cap [0,1] \to \mathbb{R}$, with $Q(0) = Q(1) = 0$, such that

$$\widehat{q}^{(n)}_i = Q(i/n) - Q((i-1)/n).$$

Conversely, every such function Q generates a family satisfying the displayed replication identity.

Proof. For each population size n, define cumulative weights on the population-n grid by $Q_n(k/n) = \sum_{i=1}^{k} \widehat{q}^{(n)}_i$ for $k \in \{0, \dots, n\}$. We first show that these grid-specific definitions agree whenever two grids contain the same population share.

Suppose $k/n = \ell/s$, and choose a common multiple N of n and s. The replication identity implies that the cumulative weight through rank k at population n equals the cumulative weight through rank kN/n at population N. Similarly, the cumulative weight through rank ℓ at population s equals the cumulative weight through rank $\ell N/s$ at population N. Since $kN/n = \ell N/s$, the two cumulative weights coincide. Thus the functions Q_n define one function Q on $\mathbb{Q} \cap [0,1]$.

By construction, $\widehat{q}^{(n)}_i = Q(i/n) - Q((i-1)/n)$. Therefore, the slope between the adjacent population-n grid points $(i-1)/n$ and i/n is $n\widehat{q}^{(n)}_i$. Because the weights are decreasing, these successive slopes are decreasing. Hence the piecewise-linear function joining the values of Q on each population grid is concave.

Now take rational shares $p < r < s$. Choose a common denominator so that all three shares belong to one population grid. Concavity on that grid gives $Q(r) \geq [(s-r)/(s-p)]Q(p) + [(r-p)/(s-p)]Q(s)$. Therefore Q is concave on $\mathbb{Q} \cap [0,1]$. The zero-sum condition gives $Q(0) = Q(1) = 0$.

Conversely, suppose that Q is concave on $\mathbb{Q} \cap [0,1]$, with $Q(0) = Q(1) = 0$, and define the weights by the stated increments. For every n, m, and i, summing the m successive increments over

the interval $((i-1)/n, i/n]$ cancels the intermediate terms and leaves $Q(i/n) - Q((i-1)/n) = \widehat{q}_i^{(n)}$. Hence the replication identity holds. ∎

The rational-domain function already generates a rank-weighted functional at every finite population size. In the absolute geometry, define

$$G_{Q_A}^{A,n}(\mathbf{x}) = \sum_{i=1}^{n} \left[Q_A\left(\frac{i}{n}\right) - Q_A\left(\frac{i-1}{n}\right) \right] x_i, \qquad \mathbf{x} \in \mathbf{S}_n.$$

In the relative geometry, define

$$G_{Q_R}^{R,n}(\mathbf{x}) = \sum_{i=1}^{n} \left[Q_R\left(\frac{i}{n}\right) - Q_R\left(\frac{i-1}{n}\right) \right] \log x_i, \qquad \mathbf{x} \in \mathbf{S}_n^{++}.$$

These definitions require only the values of Q_A and Q_R at rational population shares. When the population size is clear, write $G_{Q_A}^{A}$ and $G_{Q_R}^{R}$.

For the next result, use $(I^K, \mathbf{S}_n^K, G_{Q_K}^{K,n}) = (I^A, \mathbf{S}_n, G_{Q_A}^{A,n})$ when $K = A$, and $(I^K, \mathbf{S}_n^K, G_{Q_K}^{K,n}) = (I^R, \mathbf{S}_n^{++}, G_{Q_R}^{R,n})$ when $K = R$.

Lemma 8 (Continuity of the population-share weighting function)**.** *Fix $K \in \{A, R\}$. Suppose there is a strictly increasing function ϕ_K such that $I^K(\mathbf{x}) = \phi_K(G_{Q_K}^{K,n}(\mathbf{x}))$ for every $n \geq 2$ and every $\mathbf{x} \in \mathbf{S}_n^K$. If I^K satisfies centered-quantile L^1 continuity, then Q_K extends uniquely to a continuous concave function on $[0, 1]$.*

Proof. For readability, suppress the subscript K on Q_K. Concavity gives continuity at the rational points of $(0, 1)$. It remains to establish continuity at 0 and 1. We do so by showing that the first and last grid weights converge to zero.

For each n, define the ordered zero-sum vectors

$$d_n^L = \left(-\sqrt{n} + \frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \ldots, \frac{1}{\sqrt{n}} \right) \quad \text{and} \quad d_n^U = \left(-\frac{1}{\sqrt{n}}, \ldots, -\frac{1}{\sqrt{n}}, \sqrt{n} - \frac{1}{\sqrt{n}} \right).$$

The first concentrates a negative deviation at the poorest rank, while the second concentrates a positive deviation at the richest rank. Their associated step profiles satisfy

$$\|d_n^L\|_1 = \|d_n^U\|_1 = \frac{2}{\sqrt{n}} - \frac{2}{n^{3/2}} \longrightarrow 0.$$

We now realize these vectors as centered profiles of admissible distributions. In the absolute geometry, set $\mathbf{x}_n^L = \sqrt{n}\,\mathbf{1} + d_n^L$ and $\mathbf{x}_n^U = \sqrt{n}\,\mathbf{1} + d_n^U$. These distributions are ordered and nonnegative. Since d_n^L and d_n^U have zero mean, $C_n(\mathbf{x}_n^L) = d_n^L$ and $C_n(\mathbf{x}_n^U) = d_n^U$.

In the relative geometry, define

$$\mathbf{x}_n^L = \left(e^{-\sqrt{n}+1/\sqrt{n}}, e^{1/\sqrt{n}}, \ldots, e^{1/\sqrt{n}} \right) \quad \text{and} \quad \mathbf{x}_n^U = \left(e^{-1/\sqrt{n}}, \ldots, e^{-1/\sqrt{n}}, e^{\sqrt{n}-1/\sqrt{n}} \right).$$

These distributions are ordered and strictly positive. Their log-income profiles are d_n^L and d_n^U, respectively. Because both vectors have zero mean, $C_n(\log \mathbf{x}_n^L) = d_n^L$ and $C_n(\log \mathbf{x}_n^U) = d_n^U$.

Thus, in either geometry, the relevant centered profiles are d_n^L and d_n^U and converge in L^1 to the zero function. The zero function represents equality after centering: if $\mathbf{x} = c\mathbf{1}$, then $C_n(\mathbf{x}) = \mathbf{0}$ and, when $c > 0$, $C_n(\log \mathbf{x}) = \mathbf{0}$.

Let $\widehat{q}^{(n)}$ be the normalized weight vector generated by Q at population n. The fixed-population characterization gives zero-sum weights, and positive normalization preserves that property. Hence $\sum_i \widehat{q}_i^{(n)} = 0$, which gives

$$\widehat{q}^{(n)} \cdot d_n^L = -\sqrt{n}\, \widehat{q}_1^{(n)}, \qquad \widehat{q}^{(n)} \cdot d_n^U = \sqrt{n}\, \widehat{q}_n^{(n)}.$$

We first prove that $\widehat{q}_1^{(n)} \to 0$. Because the weights are decreasing and sum to zero, $\widehat{q}_1^{(n)} \geq 0$. If convergence failed, there would be $\delta > 0$ and a subsequence $\{n_j\}$ such that $\widehat{q}_1^{(n_j)} \geq \delta$. Along this subsequence, $\widehat{q}^{(n_j)} \cdot d_{n_j}^L \leq -\delta\sqrt{n_j}$, so these weighted values tend to $-\infty$.

Choose an admissible distribution $\mathbf{x}_- \in \mathbf{S}_{n_-}^K$ such that $t_- = G_{Q_K}^{K,n_-}(\mathbf{x}_-) < 0$. Such a distribution exists because the representing functional is nonzero on the zero-sum transformed-coordinate space. For all sufficiently large j, $G_{Q_K}^{K,n_j}(\mathbf{x}_{n_j}^L) = \widehat{q}^{(n_j)} \cdot d_{n_j}^L < t_-$. Therefore, the representation hypothesis and strict monotonicity of ϕ_K give

$$I^K(\mathbf{x}_{n_j}^L) < \phi_K(t_-) < \phi_K(0).$$

But the centered profiles of $\mathbf{x}_{n_j}^L$ converge in L^1 to zero. Centered-quantile continuity requires $I^K(\mathbf{x}_{n_j}^L) \to \phi_K(0)$, a contradiction. Hence $\widehat{q}_1^{(n)} \to 0$.

The argument for the last weight is analogous. Since the weights are decreasing and sum to zero, $\widehat{q}_n^{(n)} \leq 0$. If these weights failed to converge to zero, some $\delta > 0$ and subsequence $\{n_j\}$ would satisfy $\widehat{q}_n^{(n_j)} \leq -\delta$. Then $\widehat{q}^{(n_j)} \cdot d_{n_j}^U \leq -\delta\sqrt{n_j}$, so eventually $G_{Q_K}^{K,n_j}(\mathbf{x}_{n_j}^U) < t_-$. Strict monotonicity of ϕ_K would again keep $I^K(\mathbf{x}_{n_j}^U)$ strictly below $\phi_K(0)$, contradicting centered-quantile continuity. Therefore $\widehat{q}_n^{(n)} \to 0$.

Since $Q(1/n) - Q(0) = \widehat{q}_1^{(n)}$ and $Q(1) - Q((n-1)/n) = \widehat{q}_n^{(n)}$, the rational-domain function is continuous at both endpoints. Together with its interior continuity, this gives a unique continuous extension $\overline{Q} : [0,1] \to \mathbb{R}$.

It remains to verify that the extension is concave. For arbitrary $x, y, \alpha \in [0,1]$, choose rational sequences $x_j \to x$, $y_j \to y$, and $\alpha_j \to \alpha$. Concavity on the rational shares gives $Q(\alpha_j x_j + (1-\alpha_j)y_j) \geq \alpha_j Q(x_j) + (1-\alpha_j)Q(y_j)$. Passing to the limit gives the corresponding inequality for $\overline{Q}$. Hence the extension is concave. We henceforth write Q for this continuous extension. ∎

6.4 Variable-population characterizations

We can now state the final results. The preceding arguments begin with the rank weights obtained separately at each population size, connect and normalize them under replication, and express them as the increments of a continuous concave function on population shares. We now identify the resulting global static rankings and all dynamic indices satisfying the variable-population axioms.

For $K \in \{A, R\}$, let $\mathcal{E}_K$ consist of the centered effect profiles of admissible transitions represented at a common population size, with profiles related by further common replication treated as the

same effect. These are centered income-effect profiles when $K = A$ and centered log-income-effect profiles when $K = R$.

For an admissible transition $(\mathbf{x}, \mathbf{y})$, let $e_K(\mathbf{x}, \mathbf{y}) \in \mathcal{E}_K$ denote the effect represented by $\widetilde{\boldsymbol{\tau}}_N^K(\mathbf{x}, \mathbf{y})$ at any common multiple N of the endpoint population sizes. Further common replication does not change this element of $\mathcal{E}_K$, so $e_K(\mathbf{x}, \mathbf{y})$ is well defined.

If $\mathbf{z} = (z_1, \ldots, z_N) \in \mathcal{E}_K$ is represented at population N, define

$$G_{Q_K}^{K,N}(\mathbf{z}) = \sum_{i=1}^{N} \left[Q_K\left(\frac{i}{N}\right) - Q_K\left(\frac{i-1}{N}\right) \right] z_i.$$

The replication identity makes this value unchanged under further common replication. We therefore write $G_{Q_K}^K(\mathbf{z})$ without specifying the representing population size.

Theorem 4 (Variable-population absolute characterization)**.** *Let $I^A : \mathbf{S} \to \mathbb{R}$ satisfy centered-quantile L^1 continuity, and suppose that its restriction to each $\mathbf{S}_n$ is continuous, nonconstant, Schur concave, and translation invariant. The following statements are equivalent.*

1. *There exists a dynamic index $J^A : \mathcal{T}^A \to \mathbb{R}$ satisfying N, SC, A-UCTPI, and DPP.*
2. *There are a nonzero continuous concave function $Q_A : [0,1] \to \mathbb{R}$, with $Q_A(0) = Q_A(1) = 0$, and a strictly increasing function ϕ_A, defined on the range of $G_{Q_A}^A$, such that*

$$I^A(\mathbf{x}) = \phi_A\big(G_{Q_A}^A(\mathbf{x})\big), \qquad \text{for every } \mathbf{x} \in \mathbf{S}.$$

When these conditions hold, every dynamic index satisfying the four axioms has the form

$$J^A(\mathbf{x}, \mathbf{y}) = F_A\big(e_A(\mathbf{x}, \mathbf{y})\big),$$

where $F_A : \mathcal{E}_A \to \mathbb{R}$ satisfies $F_A(\mathbf{0}) = 0$ and $F_A(\mathbf{z}) \geq 0 \Leftrightarrow G_{Q_A}^A(\mathbf{z}) \geq 0$ for every $\mathbf{z} \in \mathcal{E}_A$. Conversely, every such function F_A defines a dynamic index satisfying N, SC, A-UCTPI, and DPP.

The function Q_A is unique up to a positive scalar. The linear choice $F_A = G_{Q_A}^A$ gives

$$J_0^A(\mathbf{x}, \mathbf{y}) = G_{Q_A}^A(\mathbf{y}) - G_{Q_A}^A(\mathbf{x}),$$

where the endpoints may be evaluated at any common population size. This extension preserves unrestricted absolute Dynamic Lorenz dominance.

Proof. We first prove that item 1 implies item 2 and identify the form of every admissible dynamic index.

At each population size, Corollary 3 gives a nonzero decreasing zero-sum weight vector, determined up to positive rescaling. By Lemma 5, DPP makes these vectors agree proportionally under replication. By Lemma 6, they can be normalized so that the weights assigned to the finer ranks replacing an original rank sum to its original weight. By Lemma 7, the normalized vectors are generated by a concave function Q_A on the rational population shares.

We next establish one ordinal representation across all population sizes. Let $\mathbf{x} \in \mathbf{S}_n$ and $\mathbf{y} \in \mathbf{S}_m$, choose a common multiple N of n and m, and let $\widetilde{\mathbf{x}} = \mathbf{x}^{[N/n]}$ and $\widetilde{\mathbf{y}} = \mathbf{y}^{[N/m]}$ be their N-replicas. By

Lemma 4, $I^A(\mathbf{x}) = I^A(\widetilde{\mathbf{x}})$ and $I^A(\mathbf{y}) = I^A(\widetilde{\mathbf{y}})$. The fixed-population representation at population N therefore gives

$$I^A(\mathbf{y}) \geq I^A(\mathbf{x}) \Leftrightarrow G^{A,N}_{Q_A}(\widetilde{\mathbf{y}}) \geq G^{A,N}_{Q_A}(\widetilde{\mathbf{x}}).$$

The replication identity gives $G^{A,N}_{Q_A}(\widetilde{\mathbf{x}}) = G^{A,n}_{Q_A}(\mathbf{x})$ and $G^{A,N}_{Q_A}(\widetilde{\mathbf{y}}) = G^{A,m}_{Q_A}(\mathbf{y})$. Hence I^A and the rational-grid functional $G^A_{Q_A}$ induce the same ordering on $\mathbf{S}$. A single strictly increasing function ϕ_A therefore satisfies $I^A = \phi_A \circ G^A_{Q_A}$.

The hypotheses of Lemma 8 now hold. Centered-quantile continuity extends Q_A uniquely to a continuous concave function on $[0, 1]$. This proves item 2.

We now recover the form of the given dynamic index. For each $\mathbf{z} \in \mathcal{E}_A$, choose an admissible transition $(\mathbf{x}, \mathbf{y})$ whose centered effect, at some common population size, represents $\mathbf{z}$, and define $F_A(\mathbf{z}) = J^A(\mathbf{x}, \mathbf{y})$. We must show that this definition is independent of the chosen transition and common population size. Suppose that another transition $(\mathbf{x}', \mathbf{y}')$ also represents $\mathbf{z}$. Replicate the endpoints of both transitions to a common population size. By the definition of $\mathcal{E}_A$, their centered effect profiles then coincide. A-UCTPI assigns the two replicated transitions the same value, while DPP identifies each replicated transition with its original transition. Hence $J^A(\mathbf{x}, \mathbf{y}) = J^A(\mathbf{x}', \mathbf{y}')$, so F_A is well defined and

$$J^A(\mathbf{x}, \mathbf{y}) = F_A\big(e_A(\mathbf{x}, \mathbf{y})\big)$$

for every admissible transition $(\mathbf{x}, \mathbf{y})$.

Normalization gives $F_A(\mathbf{0}) = 0$. If $\mathbf{z} \in \mathcal{E}_A$ is represented by a transition $(\mathbf{x}, \mathbf{y})$, then SC and the static representation give

$$F_A(\mathbf{z}) \geq 0 \Leftrightarrow I^A(\mathbf{y}) \geq I^A(\mathbf{x}) \Leftrightarrow G^A_{Q_A}(\mathbf{z}) \geq 0.$$

Thus every admissible dynamic index has the stated form.

Conversely, suppose item 2 holds and let $F_A : \mathcal{E}_A \to \mathbb{R}$ satisfy the conditions in the theorem. Define $J^A(\mathbf{x}, \mathbf{y}) = F_A(e_A(\mathbf{x}, \mathbf{y}))$. Because e_A is unchanged under common refinement and independent endpoint replication, J^A satisfies A-UCTPI and DPP. Normalization follows from $F_A(\mathbf{0}) = 0$. Moreover,

$$J^A(\mathbf{x}, \mathbf{y}) \geq 0 \Leftrightarrow G^A_{Q_A}(\mathbf{y}) \geq G^A_{Q_A}(\mathbf{x}) \Leftrightarrow I^A(\mathbf{y}) \geq I^A(\mathbf{x}),$$

so SC holds.

For the choice $F_A = G^A_{Q_A}$, the dynamic value is the difference between the weighted endpoint values. At any common population size N, summation by parts gives

$$J^A_0(\mathbf{x}, \mathbf{y}) = N \sum_{k=1}^{N-1} C^A_N\left(\frac{k}{N}; \mathbf{x}, \mathbf{y}\right)\left(\widehat{q}^{(N)}_{A,k} - \widehat{q}^{(N)}_{A,k+1}\right).$$

The coefficients are nonnegative because the weights are decreasing. Therefore, pointwise dominance of the normalized cumulative-effect curves implies a weakly larger value of J^A_0. ■

Theorem 5 (Variable-population relative characterization)**.** *Let $I^R : \mathbf{S}^{++} \to \mathbb{R}$ satisfy centered-quantile L^1 continuity in log-income coordinates. Suppose that its restriction to each $\mathbf{S}^{++}_n$ is continuous, nonconstant, and scale invariant, and that its log-coordinate representation is Schur concave. The following statements are equivalent.*

1. *There exists a dynamic index* $J^R : \mathcal{T}^R \to \mathbb{R}$ *satisfying N, SC, R-UCTPI, and DPP.*

2. *There are a nonzero continuous concave function* $Q_R : [0,1] \to \mathbb{R}$*, with* $Q_R(0) = Q_R(1) = 0$*, and a strictly increasing function* ϕ_R*, defined on the range of* $G^R_{Q_R}$*, such that*

$$I^R(\mathbf{x}) = \phi_R\big(G^R_{Q_R}(\mathbf{x})\big), \qquad \text{for every } \mathbf{x} \in \mathbf{S}^{++}.$$

When these conditions hold, every dynamic index satisfying the four axioms has the form

$$J^R(\mathbf{x}, \mathbf{y}) = F_R\big(e_R(\mathbf{x}, \mathbf{y})\big),$$

where $F_R : \mathcal{E}_R \to \mathbb{R}$ *satisfies* $F_R(\mathbf{0}) = 0$ *and* $F_R(\mathbf{z}) \geq 0 \Leftrightarrow G^R_{Q_R}(\mathbf{z}) \geq 0$ *for every* $\mathbf{z} \in \mathcal{E}_R$*. Conversely, every such function* F_R *defines a dynamic index satisfying N, SC, R-UCTPI, and DPP.*

The function Q_R *is unique up to a positive scalar. The linear choice* $F_R = G^R_{Q_R}$ *gives*

$$J^R_0(\mathbf{x}, \mathbf{y}) = G^R_{Q_R}(\mathbf{y}) - G^R_{Q_R}(\mathbf{x}),$$

where the endpoints may be evaluated at any common population size. This extension preserves unrestricted relative Dynamic Lorenz dominance.

Proof. Apply the proof of Theorem 4 in log-income coordinates. Coordinatewise logarithms preserve rank order and commute with replication, convert scale invariance into translation invariance, and transform centered relative effects into centered additive effects. The construction of Q_R, its continuous concave extension, the global representation through ϕ_R, and the characterization of the dynamic class follow exactly as in the absolute case. For the linear choice, the summation-by-parts calculation in the proof of Theorem 4, applied to the centered log-effect profile, has nonnegative coefficients because the log-rank weights are decreasing. Hence the linear extension preserves unrestricted relative Dynamic Lorenz dominance. ■

The two theorems characterize all dynamic extensions satisfying the variable-population axioms, not only those that preserve unrestricted Dynamic Lorenz comparisons. Within the characterized class, an index $J^K(\mathbf{x}, \mathbf{y}) = F_K(e_K(\mathbf{x}, \mathbf{y}))$ is weakly monotone under unrestricted K-Dynamic Lorenz dominance if and only if F_K preserves the pointwise order of the normalized cumulative-effect curves. It is unrestricted K-Dynamic Lorenz consistent if and only if F_K also assigns a strictly larger value whenever the pointwise inequalities hold at every population share and are strict at some share.

Taking $F_K = G^K_{Q_K}$ yields a linear extension that is weakly monotone. Indeed, concavity of Q_K makes its increments weakly decreasing on every finite population grid, so the summation-by-parts coefficients are nonnegative. If these increments are strictly decreasing on every finite population grid, then every strict unrestricted K-Dynamic Lorenz comparison produces a strictly larger value, and the linear extension is unrestricted K-Dynamic Lorenz consistent. Concavity alone permits equal adjacent increments and therefore guarantees only weak monotonicity.

7 Discussion

Baseline independence is the main identifying restriction in the paper. Together with Static-Change Consistency, it requires a feasible transformed-coordinate displacement to have the same direction

of static change wherever it can be implemented. The initial distribution may determine whether the displacement is feasible, but not whether it is improving or worsening.

This restriction forces the improving displacements to form a half-space and thereby yields the affine representation. The argument is established directly on the maintained economic domain, without first extending the static ranking to a larger vector space.

Schur concavity plays a different role. It orders the coefficients of the affine representation but does not produce affinity itself. As the counterexample in Section 3.1 shows, even a continuous and strictly Schur-concave ranking need not admit a baseline-independent dynamic extension.

The transformed-coordinate formulation unifies the absolute and relative results. Identity coordinates treat common additive changes as distributively neutral, whereas logarithmic coordinates treat common proportional changes as neutral. Centering removes the corresponding transformed aggregate. The absolute and relative geometries are therefore applications of the same representation principle rather than unrelated characterizations. The logarithmic formulation requires strictly positive endpoint incomes, although the associated log-income changes may be positive, zero, or negative.

The maintained axioms characterize more than the static rankings that admit an extension. Conditional on each admissible static ranking, they also identify the full class of baseline-independent dynamic indices. Such an index may be any function of the relevant effect profile whose nonnegative region agrees with that of the corresponding linear functional. The axioms determine how each transition is evaluated relative to no change, but they do not determine how arbitrary pairs of transitions are ranked.

Dynamic Lorenz monotonicity adds the requirement that the profile functional preserve the cumulative-effect order. Within the admissible class, the Dynamic-Lorenz-consistent indices are exactly those that preserve both the weak and strict parts of this order. Taking the difference between the corresponding generalized-Gini values yields a linear extension that preserves weak comparisons when the rank weights are decreasing and strict comparisons when they are strictly decreasing.

The variable-population results add a separate form of coherence. Independent-endpoint DPP first implies static replication invariance and then makes the fixed-population weight vectors proportionally consistent under replication. The proportionality factors are consistent across successive replications, which allows the weight vectors to be rescaled so that the weights assigned to the finer ranks replacing an original rank sum exactly to its original weight. Cumulative weights then define one concave function on the rational population shares, and centered-quantile L^1 continuity extends that function continuously to $[0, 1]$. The consistency of the rank weights across population sizes is derived from dynamic extendability and independent endpoint replication rather than imposed through a primitive static population principle.

The framework is deliberately anonymous and distributional. It compares ordered endpoint distributions and the rank-based effect profiles connecting them, not the trajectories of identified individuals. It does not evaluate reranking, positional mobility, or income instability at the individual level. Two transitions with the same ordered endpoint distributions are indistinguishable here even if the underlying individual trajectories differ.

Baseline independence should likewise be understood as a benchmark, not as a universally compelling requirement. If the distributive value of a reform is allowed to depend on initial inequality, the same effect profile may receive different evaluations at different baselines, the set of improving directions may vary with the initial distribution, and the present affine characterization need not follow. A baseline-sensitive theory would require additional structure to govern the trade-off between

the distributive effect of a reform and the normative relevance of its initial state.

8 Conclusion

This paper characterizes the static inequality rankings that admit baseline-independent dynamic indices. Such an index evaluates a transition as weakly improving exactly when the static ranking places the final distribution weakly above the initial distribution. For a fixed population and fixed transformed aggregate, a continuous, nonconstant, and Schur-concave static ranking admits a normalized, baseline-independent dynamic index satisfying Static-Change Consistency if and only if it is a strictly increasing transformation of a generalized-Gini functional. When the transformed aggregate may vary, translation invariance in transformed coordinates selects a zero-sum representative of the rank weights. Identity coordinates yield the absolute characterization, while logarithmic coordinates yield its relative counterpart on strictly positive incomes.

Conditional on each admissible static ranking, the paper also characterizes the complete class of baseline-independent dynamic extensions permitted by the maintained axioms. Such an extension may be any function of the relevant effect profile whose nonnegative region agrees with that of the corresponding linear functional. Within this class, the indices that preserve Dynamic Lorenz comparisons are exactly those whose profile functionals preserve the corresponding cumulative-effect order. Taking the difference between the generalized-Gini values yields a linear extension with this property; it also preserves every strict comparison when the rank weights are strictly decreasing. Thus the existence of a baseline-independent dynamic extension determines the ordinal geometry of the static ranking without requiring the dynamic index itself to be a cardinal difference of state values.

Finally, independent endpoint replication links the fixed-population representations across population sizes. After the weight vectors are consistently rescaled, the finite-population rank weights are the increments of a single continuous concave function on population shares. The extension principle characterizes the static rankings that are coherent across populations.

It also characterizes their admissible dynamic extensions. Each such index depends only on the transition's centered effect profile and is unchanged by replication of either endpoint. It evaluates a transition as weakly improving exactly when the associated weighted change is nonnegative. Within this class, the indices that preserve unrestricted Dynamic Lorenz comparisons are exactly those whose profile functionals preserve the pointwise order of the normalized cumulative-effect curves. Taking the difference between the corresponding weighted endpoint values yields a linear extension with this property.

The results concern anonymous changes between ordered distributions, not individual mobility. Moreover, the relative characterizations exclude zero-income endpoints because logarithmic coordinates are not finite at zero. Within these boundaries, the analysis provides a unified account of when ordinal static inequality rankings admit baseline-independent dynamic evaluations in absolute and relative geometries, with fixed or variable populations.

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